



PRODUCER GUIDE

RatioBos 1.0.0

The *cheapest* ration for your cattle, from the pen to the profit

A practical guide to formulating, mixing, costing and measuring the footprint of your feedlot ration with RatioBos, explained through one real case from Jalisco, start to finish.

1st edition · October 2026 · covers RatioBos 1.0.0

MIT License

Héctor Tecumshé Mojica-Zárte · Luis Ángel Barrera-Guzmán

There is a companion guide for research and teaching: the Scientist Guide.

RatioBos

Feedlot finishing · West and Bajío · \$ 4,042/t

ES EN **Producer** Scientist

Progress: 7 of 11 blocks

✓ This is the cheapest possible ration with your ingredients and prices, and it covers what your cattle need for the target you set.

Cost per t DM **\$ 4,513**
AF: \$ 4,042/t

Cost per head per day **\$ 35.00**
7.75 kg MS · 8.7 kg TCO

Expected gain **1.35 kg/d**
target 1.35 · limit: energy

Feed cost / kg gained **\$ 25.93**
feed: gain 5.74

What it means for your ranch

- The ration costs **\$ 4,513** per tonne of dry matter, i.e. **\$ 4,042** per tonne as mixed (90 % DM).
- Each animal eats about **7.8 kg of dry matter** (8.7 kg as fed) and costs you **\$ 35.00** per day.
- This ration is expected to give **1.35 kg/d** (your target: 1.35); each kilo gained costs **\$ 25.93** in feed, below your sale price of \$ 58.00/kg ✓.
- At that rate, reaching sale weight takes about **104 days** (refine it in the Program).
- For your 300 head prepare **2,598 kg** of mix per day (0.9 loads of the 3,000 kg wagon).
- What makes this ration most expensive is meeting **net energy for gain (NEg)**.
- If you can buy **Corn silage** at \$ 1.19 per kg or less, reformulate: it would start to pay off.
- If you can buy **Dry-rolled corn** at \$ 4.12 per kg or less, reformulate: it would start to pay off.

Mixing sheet · load of 3,000 kg as fed

Wet kilos, in the suggested loading order: forages and silages first, then grains and meals, and pre-mixed small ingredients last. Mix 4-6 minutes after the last load.

☐ Corn stover	354.0 kg	☐ Dry-rolled sorghum	2,500.5 kg	☐ Distillers grains with solubles (DDGS)	84.7 kg	☐ Urea (46 % N)	21.9 kg
☐ Urea (46 % N)	21.9 kg	☐ Calcium carbonate	25.0 kg	☐ Salt (NaCl)	5.52 kg	☐ Trace-mineral and vitamin premix	8.48 kg

1	2	3	4	5	6	7	8	9	10	11
Animal	Region	Requirements	Management	Ingredients	Prices	Formulation	Program	Dashboard	Sustainability	Report

P How to use this guide

Who it is for, how it is organised and what each sign on the page means.

This guide is for whoever decides what the cattle on a ranch or in a feedlot eat: the producer, the mixer-wagon operator, the veterinarian or nutritionist who advises several operations. You need no coding or maths; you need to know your cattle.

It follows the app's order: one chapter for each block in the top bar, 1 to 11, each with its own colour. Every example uses **a single case**, a 300-bull feedlot in Jalisco, so the numbers in one chapter are the ones that feed the next. Read it start to finish and you will have formulated, mixed, costed and footprinted a real ration.

Conventions

WRITTEN AS	MEANS
 Formulate	An app button or field, with its text exactly as on screen.
Report > Save project	A path: open the first item, then the next.
380	A value you type as shown.
1	A marker on a screenshot; it is explained below with the same number.
MS · TCO	Dry matter · as fed (the wet feed, as the mixer scale weighs it). The app writes DM and AF in English.

The five boxes



IN THE PEN

The worked case, step by step, with the real numbers the app returns.



TIP

Shortcuts and good practice that save time or money.



CAUTION

Common mistakes that can cost money or animal health.



WHY?

The reason behind a recommendation, in a few lines. You can skip it.



DECISION RULE

A short table: if you see this, do that.



TWO GUIDES, TWO VIEWS

RatioBos has a **producer view** and a **scientist view** (button in the top bar). This guide covers the producer view. The equations, the linear model, the validation and the research export are in the Scientist Guide.

O

Open the app and meet the screen



RatioBos is a web page that runs on its own in your browser. Nothing to install, no account, and your data goes nowhere.

0.1 Opening RatioBos

Go to hectortecumshe-ai.github.io/RatioBos from a computer, tablet or phone, or scan the QR code on the last page. The first time it downloads a single file of under half a megabyte; after that it works without internet if you leave the tab open, and you can save the page (**Ctrl** + **S**) to use it on the ranch with no signal.

0.2 The home screen

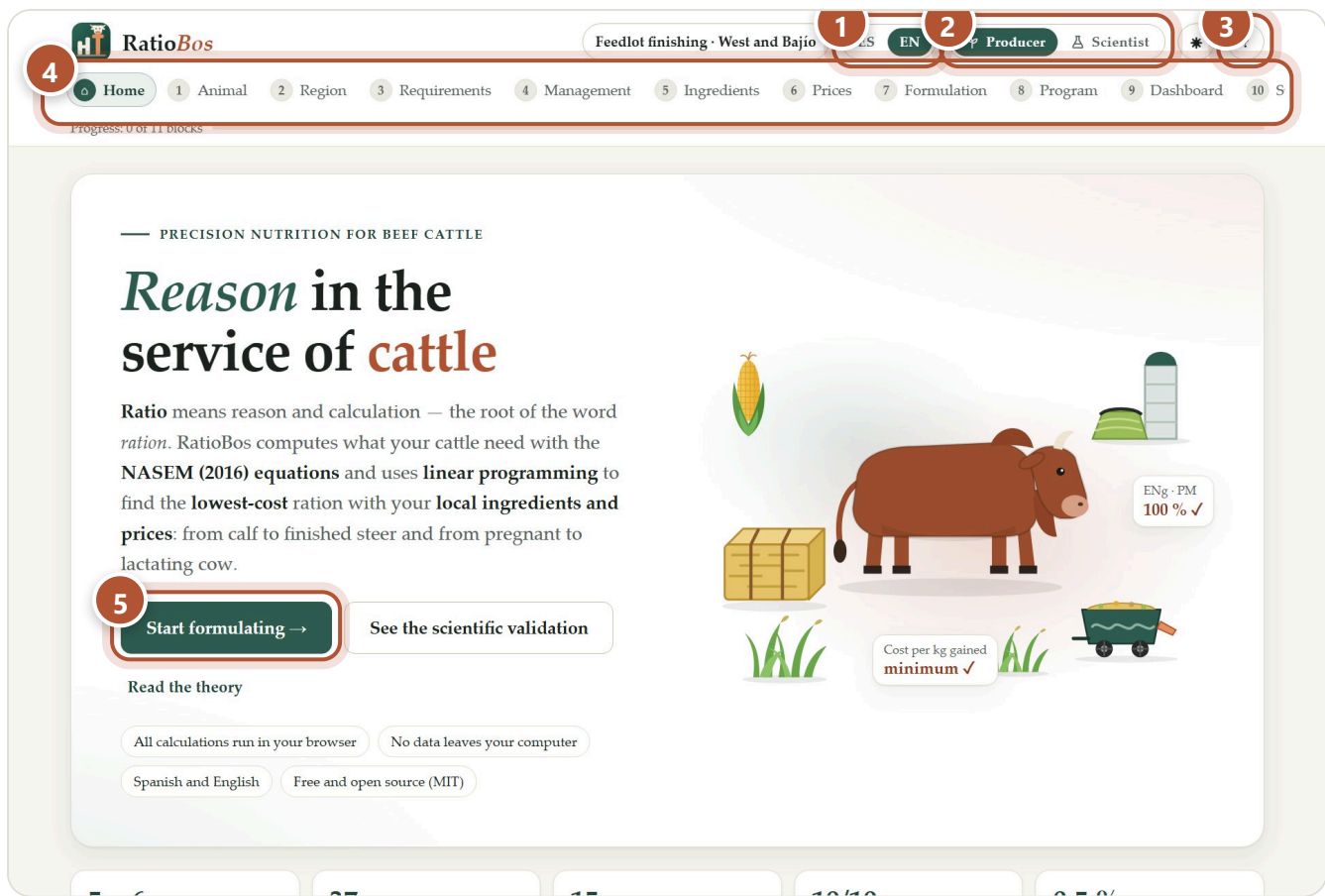


FIGURE 0.1 Home. Everything you need to get around is in the top bar.

- 1** ES / EN switches the language of the whole app, reports included.
- 2** **Producer / Scientist** switches the view; your data is kept.
- 3** ? opens the general help. Every field also has its own ?.
- 4** The **block bar**, 1 to 11. You can jump to any of them; a tick marks the ones you have visited.
- 5** **Start formulating →** takes you to Block 1.

0.3 Two views, one app

The producer view shows only what you need to decide: the 8 key nutrients, a rumen-health traffic light, the mixer sheet in wet kilos, the expected gain and the cost per kilo produced. The scientist view adds the full model. Switch at any time, for instance to show your advisor where a number comes from.

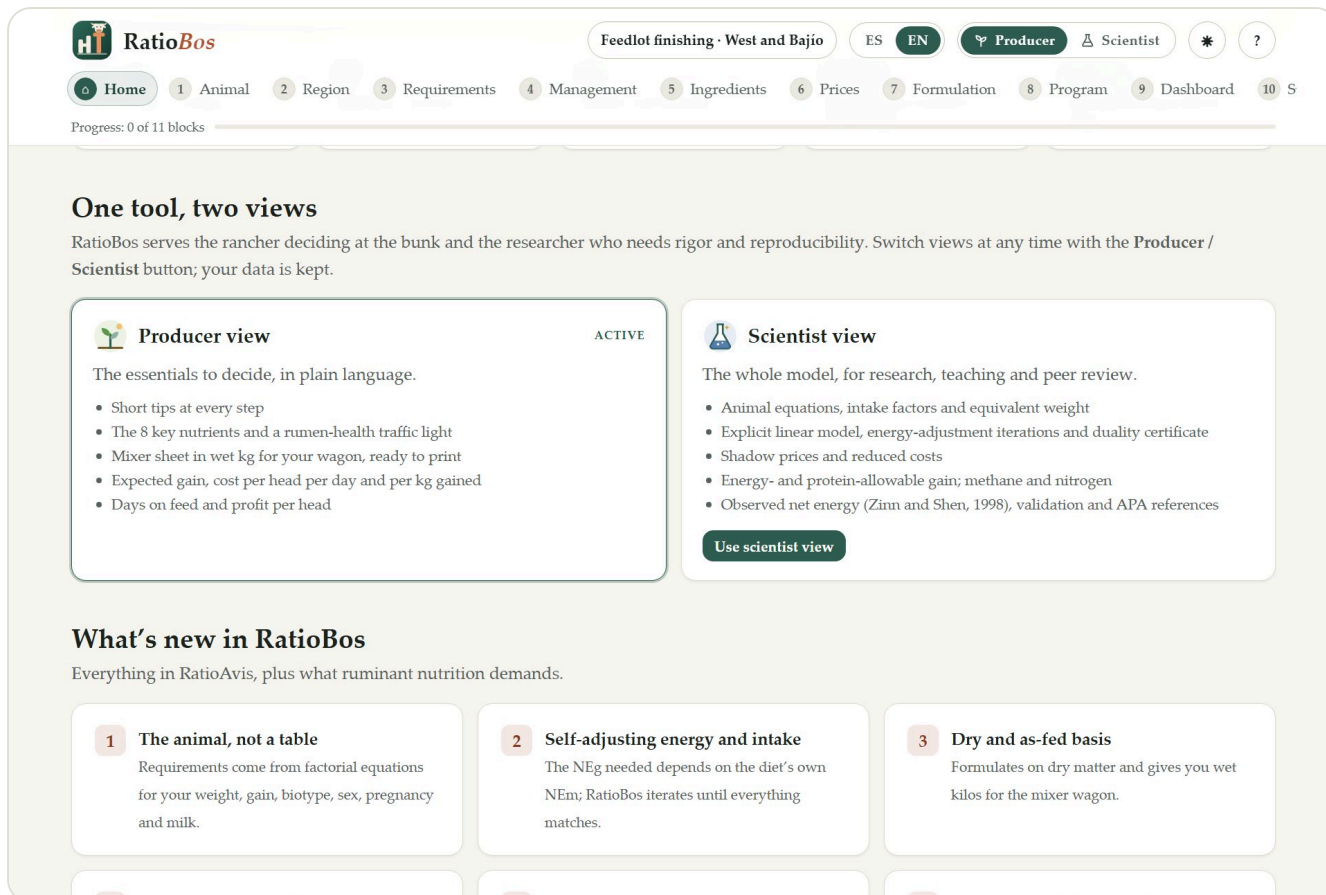


FIGURE 0.2 One tool, two views. The active card has a border.

0.4 Help for every field

Hover (or tap, on a phone) over any **?**. You get what the field means and a colour-coded **decision scale**: green, fine; amber, watch; red, fix. It is the quickest way to learn without leaving the app.

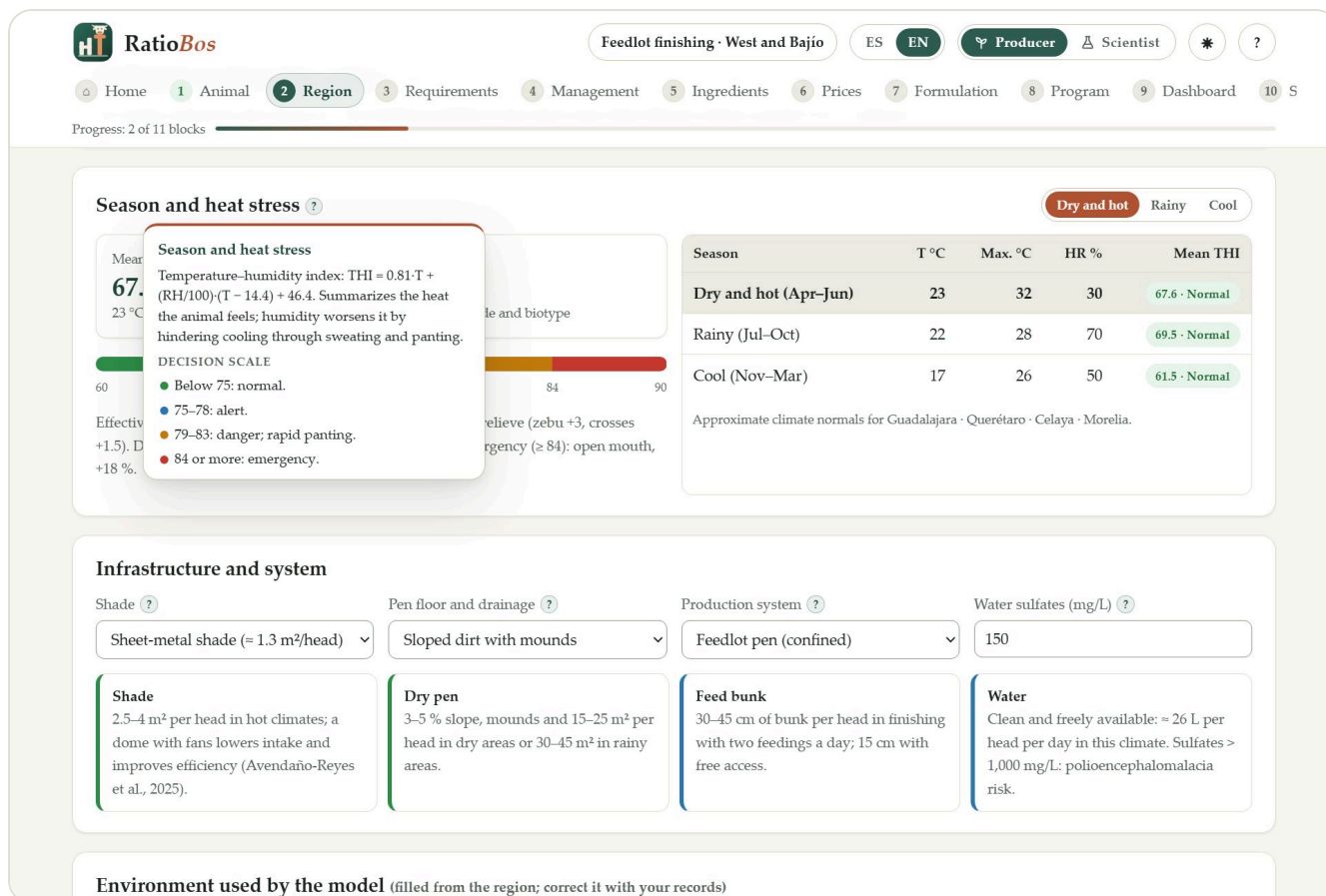


FIGURE 0.3 Help for the temperature-humidity index (THI). Each level of the scale tells you what to do.

0.5 The case in this guide

So that every step has concrete numbers, we will follow one pen from start to finish:

ANIMALS 300 intact bulls	BIOTYPE Zebu × European cross	ENTRY WEIGHT 380 kg
LOCATION Jalisco · West and Bajío	SEASON Dry and hot (Apr–Jun)	MIXER WAGON 3,000 kg load

We buy at **\$62** per kilo live weight and sell at **\$58**. The target is **1.35 kg a day** up to 520 kg. It is a typical Bajío feedlot: local sorghum and corn, corn stover for fibre and a sheet-metal shade over the pens.

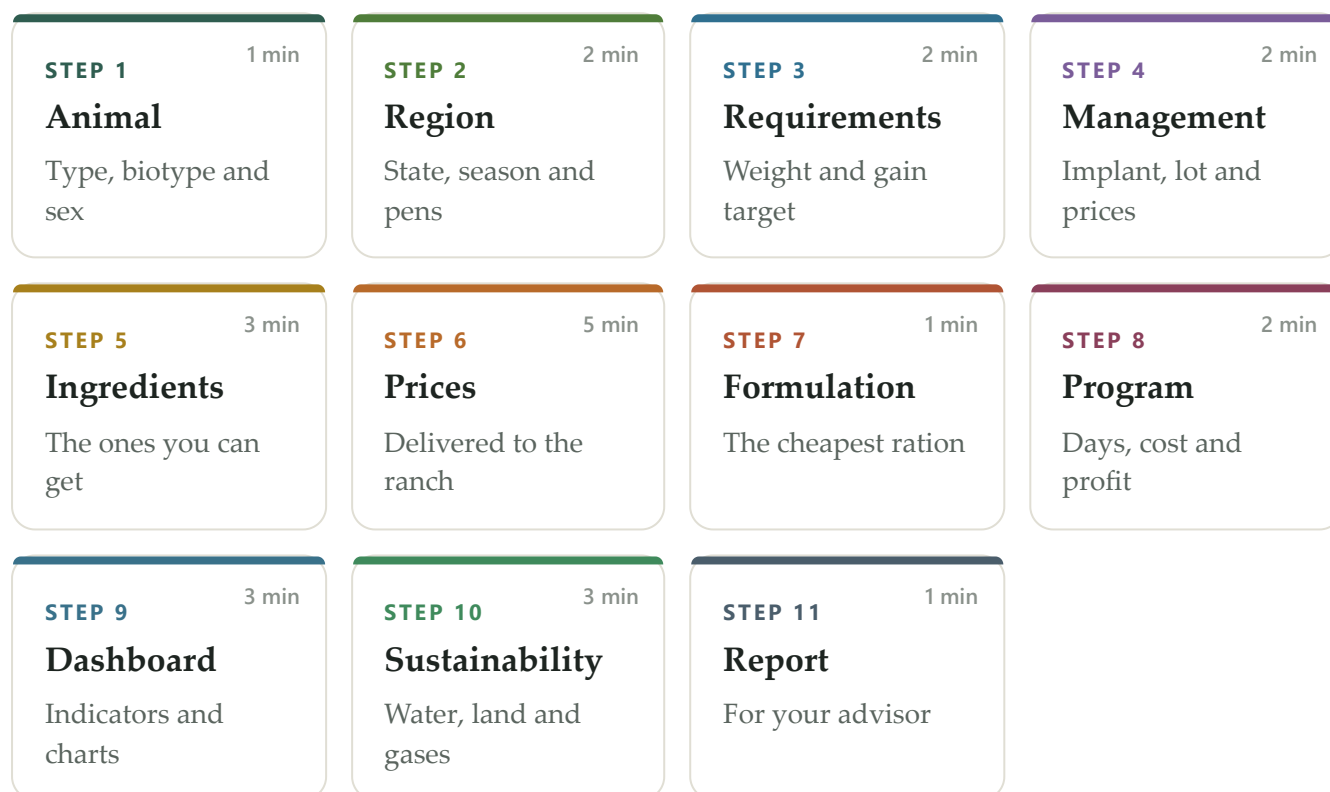


WHY SELL FOR LESS THAN YOU BUY?

A light calf costs more per kilo than a finished steer. A feedlot makes money because each kilo the animal puts on costs less in feed than it is worth at sale. That is why the number that matters most in RatioBos is the **feed cost per kilo gained**.

0.6 The path in eleven steps

The first time takes about 25 minutes. After that, with your data saved, updating prices and reformulating takes under two.



0.7 Your work saves itself

Every change is stored in that computer's browser. Close the tab and come back tomorrow, and everything is as you left it. To move it to another computer or back it up, use the path above (chapter 11).

1

Animal type, biotype and sex



First, know whom you are feeding. A calf, a finishing bull and a nursing cow need very different rations.

1.1 Choose the animal type

There are five cards. The first three use the **growth model** (the animal must gain weight); the last two, the **cow model** (maintain, carry a calf or make milk).

RatioBos Feedlot finishing - West and Bajio ES EN Producer Scientist

Home 1 Animal 2 Region 3 Requirements 4 Management 5 Ingredients 6 Prices 7 Formulation 8 Program 9 Dashboard 10 S

Progress: 1 of 11 blocks

BLOCK 1 OF 11

Animal type, biotype and sex

First, know whom you are feeding. The animal type decides which equations the model uses (growth or cow); biotype and sex change maintenance cost, mature size and what each kilo of gain carries.

IN A NUTSHELL
Choose which animals you will feed, their breed and whether they are bulls, steers or heifers. The app then knows how much they spend and can gain.

Animal type ?

Growing calves

From weaning to ~ 320 kg. Forage-rich diet to grow frame and muscle without fattening.

Growth model

1 Feedlot finishing

Bulls and steers in the feedlot up to market weight. High-grain diets: maximum gain with rumen health.

Growth model

Replacement heifers

Moderate growth to reach breeding at 60–65 % of mature weight, without fattening.

Growth model

Pregnant cows

Dry beef cows in mid or late gestation. The last trimester decides calf weight and the next pregnancy.

Cow model

Lactating cows

Cows nursing their calf. Milk sharply raises energy, protein and mineral needs.

Cow model

FIGURE 1.1 Block 1. The selected card shows a tick and a label with the model it uses.

1 Feedlot (finishing), the one in our case.

TYPE	WHAT IT IS FOR
Growing calves	Weaning to $\approx$ 320 kg. Plenty of forage: build bone and muscle without fattening.
Feedlot	Bulls and steers up to sale weight. High-grain diets: top gain with a healthy rumen.
Replacement heifers	Moderate growth to reach breeding at 60–65 % of mature weight, without getting fat.
Pregnant cows	Dry cows in mid or late pregnancy. The last third decides calf weight and the next pregnancy.
Lactating cows	Nursing cows. Milk sharply raises energy, protein and mineral needs.

1.2 Biotype and sex

The biotype groups breeds that behave alike in the pen. It changes how much the animal spends just staying alive (its **maintenance**) and at what weight it fattens. Sex does the same: an intact bull spends 15 % more on maintenance, but builds more muscle.

RatioBos Feedlot finishing - West and Bajío ES EN Producer Scientist

Home 1 **Animal** 2 Region 3 Requirements 4 Management 5 Ingredients 6 Prices 7 Formulation 8 Program 9 Dashboard 10 S

Progress: 1 of 11 blocks

Biotype ?

British

Angus, Hereford

Early maturing, fatten quickly and marble well. Reference maintenance (1.00).

Maintenance × 1.00

Continental

Charolais, Limousin, Simmental

Large mature size and heavy muscling: more days and weight to finish. Simmental and dual-purpose breeds have higher maintenance (1.20).

Maintenance × 1.00

Zebu (Bos indicus)

Brahman, Nellore, Gyr, Guzerat, Indubrasil, Sardo Negro

Tropically adapted: tolerate heat and ticks, with 10 % lower maintenance. Slower growth and less marbling.

Maintenance × 0.90

1 Zebu × European cross

F1, Beefmaster, Brangus, Simbrah, Charbray, Santa Gertrudis, Brown Swiss × zebu

The most common feedlot animal in tropical Mexico: hybrid vigor, hardiness and good gain. Intermediate maintenance (0.95).

Maintenance × 0.95

Mexican Criollo

Coreño, Chinampo, Rarámuri Criollo

Descendants of 16th-century Spanish cattle: small, hardy, tolerant of drought and poor forage. Useful on rangeland and in crossbreeding.

Maintenance × 1.00

Dairy

Holstein, Brown Swiss males

Dairy bull calves fed for beef: 20 % higher maintenance and 8 % more intake; large frame.

Maintenance × 1.20

Sex ?

Steer

Model reference.

2 Bull

15 % higher maintenance; more muscle and less fat at the same weight.

Heifer

Fattens at a lighter weight: equivalent final weight = 18 % lower.

FIGURE 1.2 Biotypes and sex. The blue label on each card is its maintenance factor.

- 1 Zebu × European cross** (F1, Beefmaster, Brangus, Simbrah...): maintenance × 0.95.
- 2 Bull (intact).**



DECISION RULE · WHICH BIOTYPE?

Angus, Hereford	British (× 1.00). Fatten early: sell lighter.
Charolais, Limousin, Simmental	Continental . More days and weight to finish.
Brahman, Nelore, Gyr, Indubrasil	Zebu (× 0.90). Handles heat; grows slower.
F1, Beefmaster, Brangus, Brown Swiss × zebu	Cross (× 0.95). The most common in Mexican feedlots.
Coreño, Chinampo, Rodeo Criollo	Mexican Criollo . Small and hardy; suited to rangeland.
Holstein, Brown Swiss males	Dairy (× 1.20). Spends more and eats 8 % more.



IN THE PEN

Our bulls are zebu × European crosses from a lot on the Jalisco coast. We tap **Feedlot**, **Zebu × European cross** and **Bull (intact)**, and move on with [Next →](#).

Region, climate and infrastructure



2

Fattening cattle in Sonora is not the same as in Querétaro or Veracruz. Climate, grain prices and pen condition change gain and cost.

2.1 Tap your state on the map

The map groups Mexico's 32 states into **9 cattle regions**. Tap your state and the app loads the region's climate, soils, breeds, production systems, inputs and challenges, and below it shows published results from that region when available.

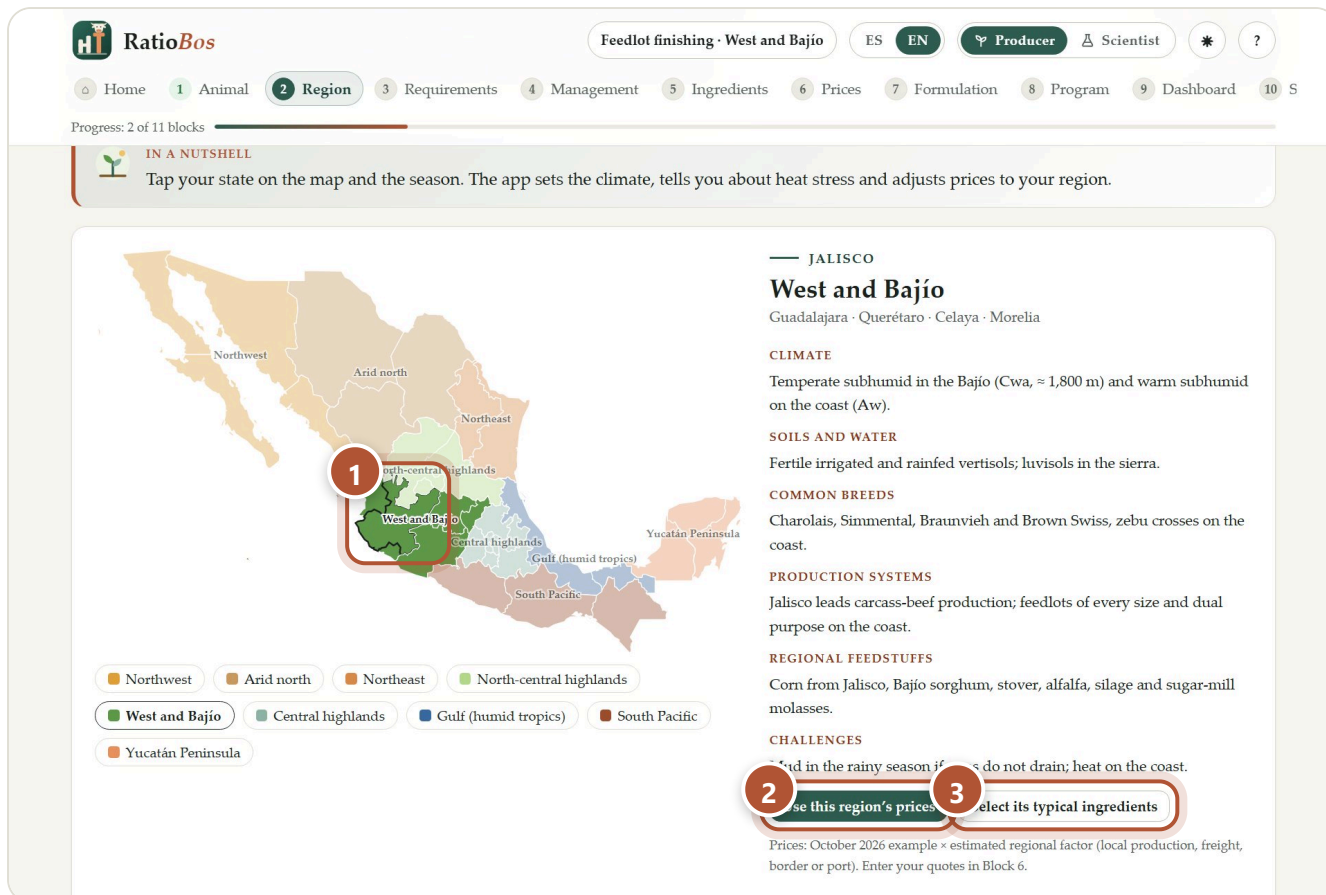


FIGURE 2.1 Block 2. Jalisco, in the West and Bajío region.

- 1 The chosen state has a black outline; its region lights up.
- 2 **Use this region's prices** adjusts the sample prices by a regional factor (local supply, freight, border or port).
- 3 **Select its typical ingredients** ticks in Block 5 the feeds common in the region.

! CAUTION · REGIONAL PRICES ARE A GUIDE

They are for comparing regions and getting started. Before mixing, enter in Block 6 what **you** pay this week, delivered to the ranch.

2.2 Season and heat stress

Pick the season: **dry and hot** (April to June), **rainy** (July to October) or **cool** (November to March). The app computes the **temperature-humidity index (THI)**, which sums up the heat the animal feels: with high humidity it is harder to cool off by sweating and panting.

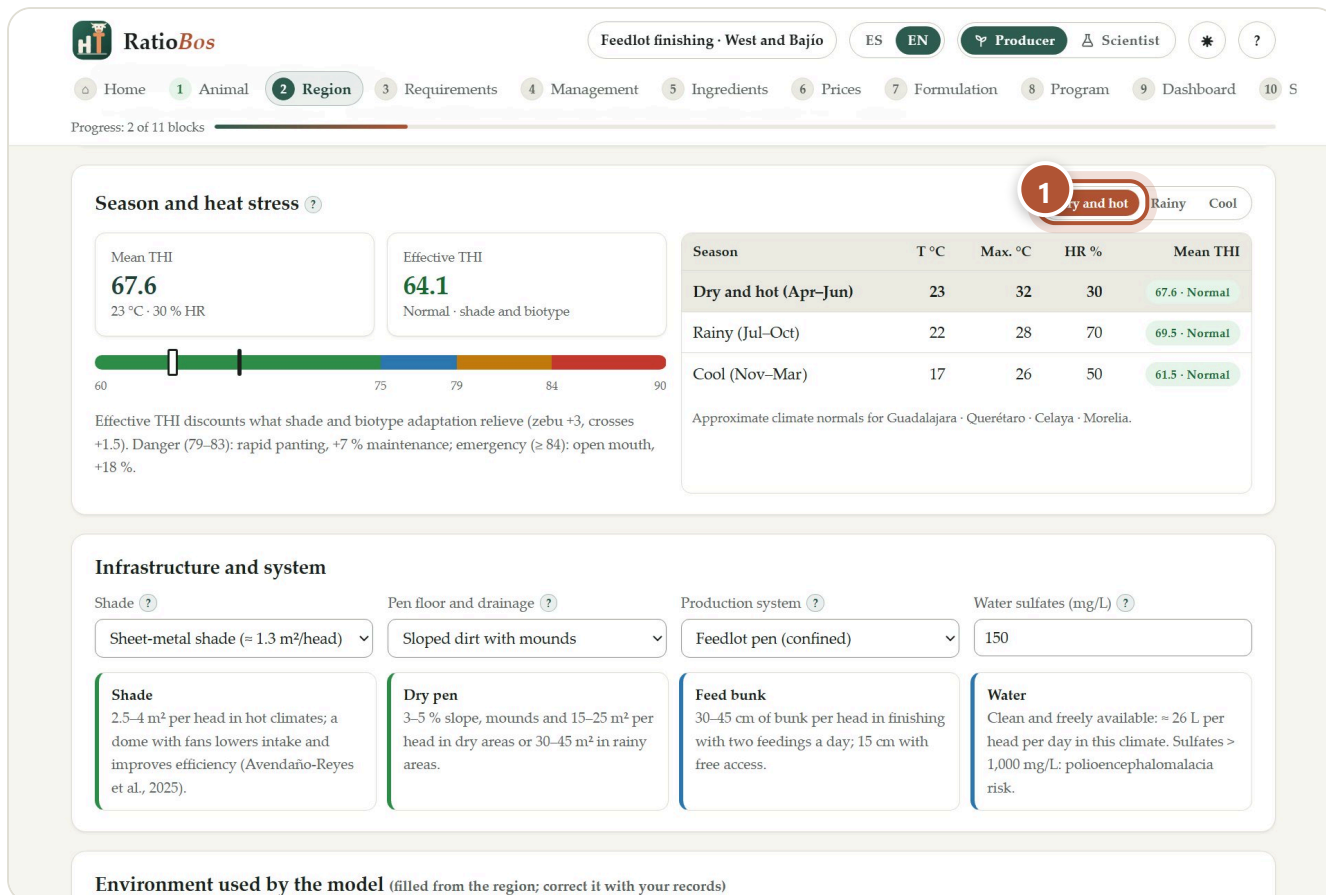


FIGURE 2.2 Season and THI. The bar runs from green (normal) to red (emergency). The black tick is your effective THI.

1 The active season, in terracotta.

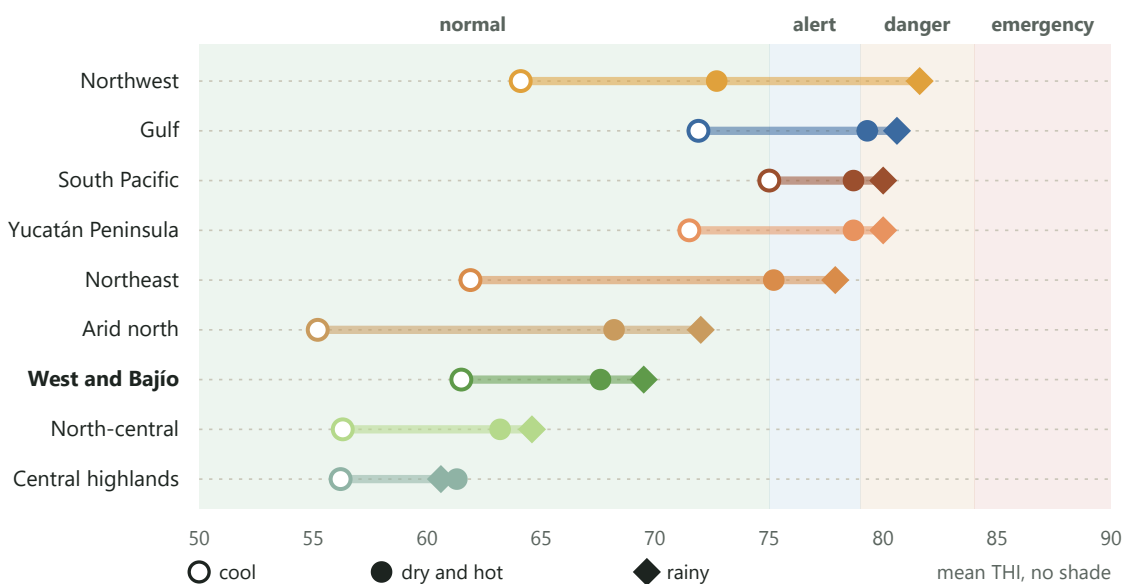


FIGURE 2.3 Mexico's heat by region and season. Mean THI for each region (before shade or biotype credit). The humid tropics and coasts cross into the alert zone in the dry and rainy seasons; the highlands hardly ever do. Data from the app.



DECISION RULE · EFFECTIVE THI

Below 75 · normal

No adjustment. Clean water, free access.

75–78 · alert

Check shade and troughs. Feed more in the late afternoon.

79–83 · danger

Rapid panting; maintenance rises 7 % and gain drops. Add shade.

84 or more · emergency

Open-mouth panting; +18 % maintenance. Sprinklers or fans; do not move cattle in the heat of the day.

2.3 Infrastructure and system

Four facts about your pens enter the model. Each has a card underneath with the practical recommendation.

Feedlot finishing - West and Bajío
ES EN
🔍 Producer
🔍 Scientist
⌵
?

Home
1 Animal
2 Region
3 Requirements
4 Management
5 Ingredients
6 Prices
7 Formulation
8 Program
9 Dashboard
10 S

Progress: 2 of 11 blocks

Infrastructure and system

1
Shade
Sheet-metal shade (≈ 1.3 m²/head)

Shade
2.5–4 m² per head in hot climates; a dome with fans lowers intake and improves efficiency (Avendaño-Reyes et al., 2025).

2
Floor and drainage
Sloped dirt with mounds

Dry pen
3–5 % slope, mounds and 15–25 m² per head in dry areas or 30–45 m² in rainy areas.

3
Production system
Feedlot pen (confined)

Feed bunk
30–45 cm of bunk per head in finishing with two feedings a day; 15 cm with free access.

4
Water sulfates (mg/L)
150

Water
Clean and freely available: ≈ 26 L per head per day in this climate. Sulfates > 1,000 mg/L: polioencephalomalacia risk.

Environment used by the model (filled from the region; correct it with your records)

Previous month mean temperature (°C)
17

Current mean temperature (°C)
23

Relative humidity (%)
30

Panting
No panting

Pen mud depth (cm)
0

Nights
☒ Cools down at night (< 20 °C)

← Animal

Next
3 · Weight, target and requirements →

FIGURE 2.4 Infrastructure and system.

- Shade:** takes points off the THI. A dome with fans helps more than a metal sheet.
- Floor and drainage:** decides how much mud there will be in the rains. Mud chills and tires the animal: it eats more and gains less.
- System:** on pasture the animal walks; on hills it spends up to 20 % more.
- Water sulfates:** water sulfur adds to the ration's. Above 1,000 mg/L there is a risk of polioencephalomalacia.

◆ IN THE PEN

We tap **Jalisco** and press **Use this region's prices**. In the **dry and hot** season the Bajío averages 23 °C and 30 % humidity: a THI of **67.6**, which drops to **64.1** with the metal shade and the cross's heat tolerance. Normal: heat does not hold our bulls back at this time of year. We keep a **sloped floor with mounds**, feedlot system and 150 mg/L sulfates.

17

Weight, target and requirements



3

From your animals' weight and what you want to achieve, the app computes their daily needs with the NASEM (2016) equations.

3.1 Weight and gain target

Enter the lot's average **current live weight** and the **daily gain target**. If you weigh your cattle regularly and know what they really eat, enter the **actual DM intake**; leave it blank and the app predicts it. For cows, instead of gain you enter the days pregnant or the litres of milk.

RatioBos Feedlot finishing · West and Bajío ES EN **Producer** Scientist * ?

Progress: 3 of 11 blocks

Home 1 Animal 2 Region 3 **Requirements** 4 Management 5 Ingredients 6 Prices 7 Formulation 8 Program 9 Dashboard 10 Settings

Feedlot finishing · Zebu × European cross · Bull NASEM (2016)

1 Current live weight (kg) ? 380 Reference final weight (kg) ? 620 at target fitness 2 Target daily gain (kg/d) ? 1.35 Actual DM intake (kg/d) ? optional; empty = predicted by the app

Daily requirements per head ?

Expected DM intake ? 7.53 kg/d 1.98 % of body weight	NEm for maintenance ? 7.20 Mcal/d keep weight	Retained energy (NEg) ? 5.46 Mcal/d 1.35 kg/d of gain	Metabolizable protein ? 720 g/d 317 maint. + 403 growth	Calcium · phosphorus ? 39.8 · 20.1 g/d 50 % and 68 % absorption
---	--	--	--	--

Concentration per kg of dry matter from the animal [Recompute from the animal](#)

Leave a cell empty for no constraint. Energy, metabolizable protein, Ca and P come from the equations; forage, fiber, fat, sulfur and sodium are practical limits for the animal type. When formulating, energy and intake self-adjust to the diet (unless you edit the table).

Producer view: the 8 key nutrients are shown. The rest are still enforced when formulating.

Nutrient	Unit	Minimum ?	Maximum
Net energy for maintenance (NEm) ?	Mcal/kg	—	—

FIGURE 3.1 Block 3. The animal's data.

- 1 Current live weight: 380 kg. 2 Daily gain target: 1.35 kg/d.



TIP · A REALISTIC TARGET

On high-grain feedlot diets, crosses gain 1.2–1.5 kg/d; zebu, 1.0–1.3; British and Continental, 1.4–1.7. Ask for more than the ingredients allow and the app will tell you the maximum feasible gain (chapter 7).

3.2 What each animal needs per day

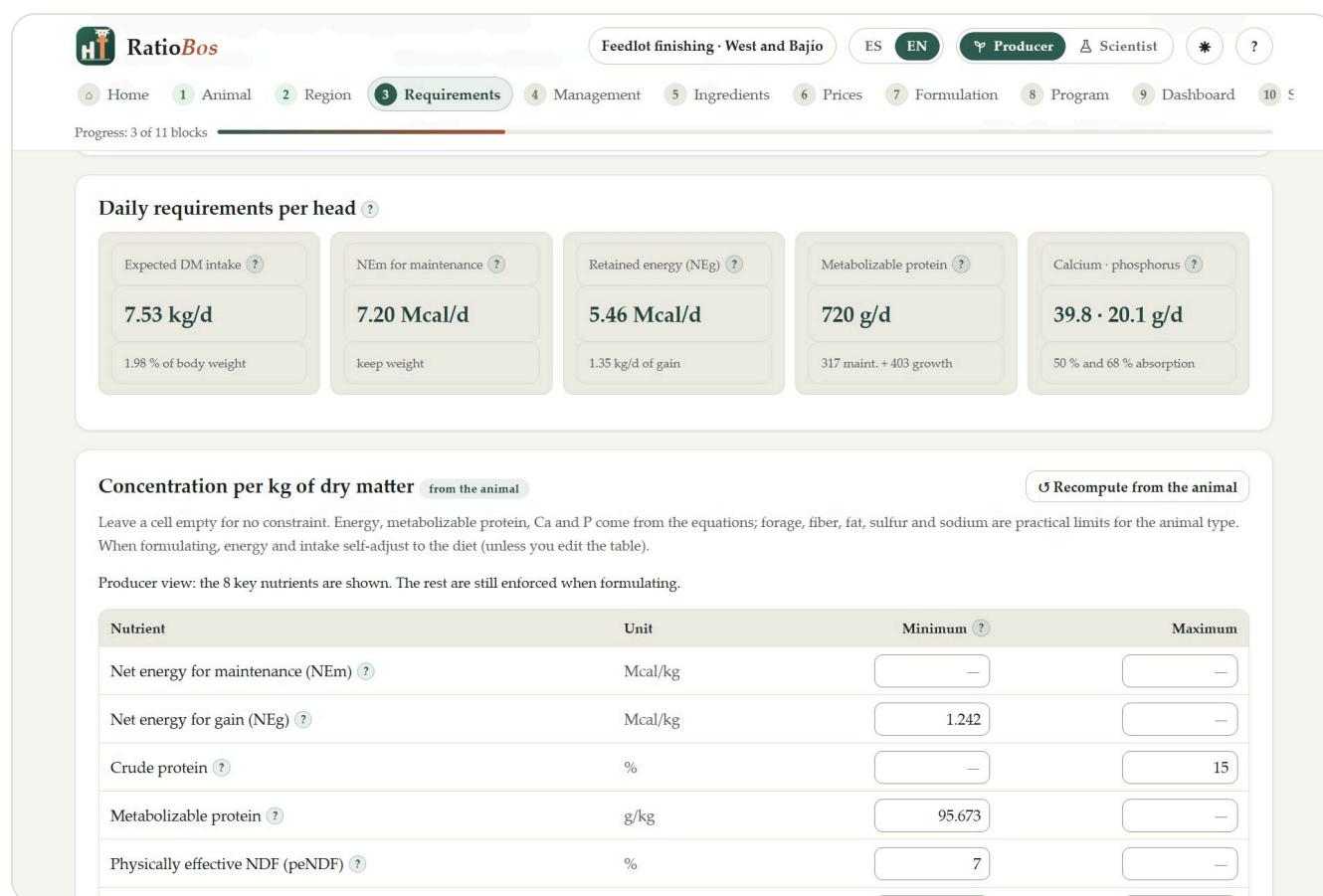
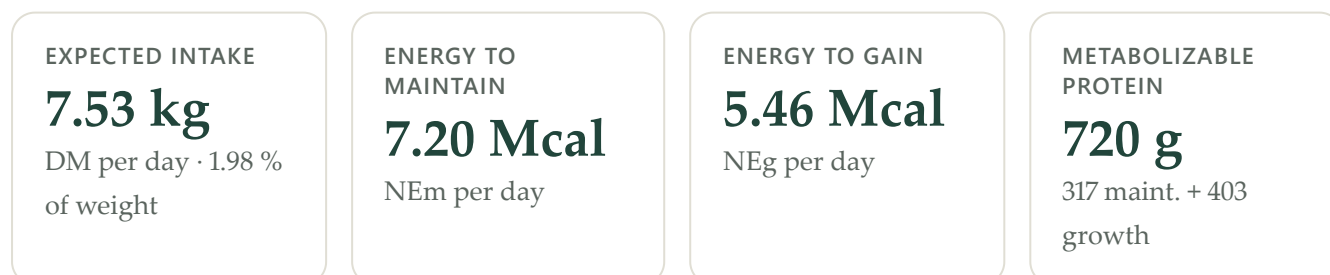
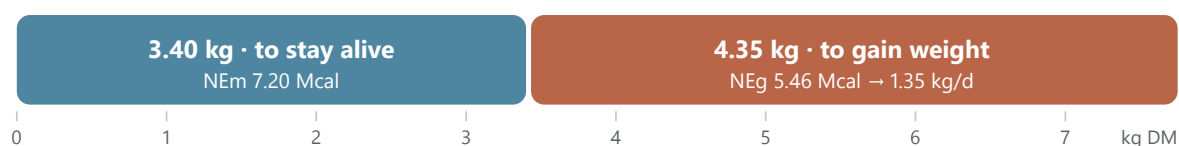


FIGURE 3.2 Daily requirements per animal for the 380 kg bull that must gain 1.35 kg/d.



What one bull eats in a day: 7.75 kg of dry matter



44 % of the feed only keeps the animal alive: that cost does not depend on how much it gains.

That is why a slow-gaining animal spreads maintenance over fewer kilos and its cost per kilo gained rises.

More gain, as far as the ration allows, makes each kilo cheaper (chapter 9).

FIGURE 3.3 Where the feed goes. Split of the case bull's intake between maintenance and gain, as computed by the app (section 7.5).



WHY TWO ENERGIES?

Cattle use feed energy with different efficiency to maintain themselves than to gain weight. That is why the California system, used by NASEM, separates **net energy for maintenance (NEm)** from **net energy for gain (NEg)**. Maintenance is covered first; whatever is left becomes kilos.

3.3 The concentration table

The daily amounts become **how much each kilo of dry matter must contain**. Energy, metabolizable protein, calcium and phosphorus come from the equations; forage, effective fibre, fat, sulfur and sodium are practical limits for the animal type. The producer view shows the 8 key nutrients; the rest are still enforced when formulating.

Feedlot finishing - West and Bajio
ES EN
Producer
Scientist
*
?

Home
1 Animal
2 Region
3 Requirements
4 Management
5 Ingredients
6 Prices
7 Formulation
8 Program
9 Dashboard
10

Progress: 3 of 11 blocks

Concentration per kg of dry matter from the animal

Leave a cell empty for no constraint. Energy, metabolizable protein, Ca and P come from the equations; forage, fiber, fat, sulfur and sodium are practical limits for the animal type. When formulating, energy and intake self-adjust to the diet (unless you edit the table).

Producer view: the 8 key nutrients are shown. The rest are still enforced when formulating.

Nutrient	Unit	Minimum ?	Maximum
Net energy for maintenance (NEm) ?	Mcal/kg	—	—
Net energy for gain (NEg) ?	Mcal/kg	1.242	—
Crude protein ?	%	—	15
Metabolizable protein ?	g/kg	95.673	—
Physically effective NDF (peNDF) ?	%	7	—
Forage in the diet ?	%	8	15
Calcium ?	%	0.529	—
Phosphorus ?	%	0.268	—

← Region
Next
4 · Management and market →

1

Recompute from the animal

FIGURE 3.4 Concentration per kg of dry matter. An empty cell does not constrain.

1 **↺ Recompute from the animal** goes back to the equation values if you edited anything.

! CAUTION · EDITING THE TABLE SWITCHES OFF THE AUTOMATIC ADJUSTMENT

The energy needed per kilo depends on the ration itself: a denser ration is eaten in smaller amounts. RatioBos adjusts energy and intake on its own until they match. If you type your own values, the app keeps yours and stops adjusting. Only do it if your nutritionist gave you a specification.

Management and market



4

Implants, ionophores and lot size change what the animal eats and yields; buying and selling prices decide whether the business makes a profit.

4.1 Management and health

RatioBos Feedlot finishing · West and Bajío ES EN Producer Scientist * ?

Progress: 4 of 11 blocks

Management

Growth promoters ?
☒ Anabolic implant

Additive ?
☒ Ionophore (monensin, lasalocid)

Head in the group ?

Mixer wagon load (kg as fed) ?

Health and additives ?

Ionophore dose (mg/kg DM) ?

In-feed antibiotic

Antibiotic dose (mg/kg DM)

β -agonist at the end

Dose (mg/kg DM) · days

Implants per head

Dewormings

Vaccines (doses)

They do not change the ration: they feed the medicine inventory in Block 10. Clenbuterol is banned in Mexico; authorized β -agonists require their withdrawal period before slaughter.

Market

Currency (symbol)

Purchase price (\$/kg live) ?

Sale price (\$/kg live) ?

Other costs (\$/head/day) ?

Comfort conditions
 No special environmental adjustments.

FIGURE 4.1 Block 4 · Management.

- 1 Anabolic implant.** Without one, expected intake drops $\approx 6\%$.
- 2 Ionophore** (monensin, lasalocid). Lowers intake $\approx 3\%$ and improves energy use: the ration's NEm is multiplied by 1.12.

Also enter how many **animals are in the lot** and the **mixer-wagon load** in as-fed kilos: from those the app works out how many kilos to mix per day. The **Health and additives** card (ionophore dose, antibiotic, β -agonist, implants, dewormings and vaccines) does not change the ration: it feeds the medicine inventory in Block 10.

! CAUTION · CLENBUTEROL IS BANNED

In Mexico clenbuterol is banned in livestock. Approved β -agonists (zilpaterol, ractopamine) require the withdrawal period before slaughter, and some export markets do not accept them.

4.2 Market

RatioBos

Feedlot finishing · West and Bajío

ES EN

Producer Scientist

Home 1 Animal 2 Region 3 Requirements 4 Management 5 Ingredients 6 Prices 7 Formulation 8 Program 9 Dashboard 10 S

Progress: 4 of 11 blocks

Dose (mg/kg DM) · days

6.5 30

Implants per head

1

Dewormings

1

Vaccines (doses)

2

They do not change the ration: they feed the medicine inventory in Block 10. Clenbuterol is banned in Mexico; authorized β -agonists require their withdrawal period before slaughter.

Market

Currency (symbol)

\$

1 Purchase price (\$/kg live) (?)

2 Sale price (\$/kg live) (?)

3 Other costs (\$/head/day) (?)

Comfort conditions

No special environmental adjustments.

← Requirements

Next 5 · Available ingredients →

RatioBos 1.0.0 · Beef cattle ration formulation · Héctor Tecumshé Mojica Zárate · Luis Ángel Barrera Guzmán · 2026

Decision-support tool: results should be reviewed by a qualified animal nutritionist · MIT

FIGURE 4.2 Market. Prices per kilo live weight.

- 1 Purchase price of the animal entering the pen.
- 2 Sale price of the finished animal.
- 3 Other costs per head per day: labour, health, power, interest on capital.

◆ IN THE PEN

We keep implant and ionophore ticked (30 mg monensin per kg DM), type 300 animals and a 3000 kg load. In the market card: buy 62 , sell 58 and 9 pesos of other costs per head per day.

5

Available ingredients



Tick only what you can actually get. The more options the app has, the cheaper the ration can be.

5.1 The ingredient library

RatioBos ships with **37 ingredients** grouped into dry forages, silages and fresh forages, grains, by-products, protein sources, non-protein nitrogen, fats, minerals and premixes. Their values come from the Oklahoma State University table (Beck, Lalman & Moehlenpah, 2024) with NASEM (2016) data, and all are on a **dry-matter basis**.

Ingredient	MS %	ENm	ENg	CP %	NDF %	Notes
DRY FORAGES						
☒ Alfalfa hay (mid-bloom)	88	1.32	0.75	18.0	46	Ca, P and S from fresh alfalfa in the same table.
☐ Oat hay	91	1.17	0.62	9.0	59	Minerals from fresh oat forage.
☐ Sudangrass hay	91	1.19	0.57	9.0	67	Boot stage.
☐ Bermudagrass or stargrass hay	90	1.08	0.53	10.0	75	Early bloom; quality drops fast with maturity.
☒ Corn stover	85	0.97	0.42	6.0	67	Cheap, highly effective fiber; poor in energy and protein.
SILAGES AND FRESH FORAGES						
☒ Corn silage	34	1.61	1.01	8.0	44	Well eared. 34 % DM: weighs almost three times more as fed.
☐ Sorghum silage	36	1.17	0.62	10.0	58	Early bloom.
☐ Oat silage	35	1.23	0.66	13.0	59	
☐ Fresh bermuda or stargrass (cut)	30	1.48	0.88	16.0	68	Vegetative. Useful for cows and growers in the tropics.
GRAINS						
☒ Dry-rolled corn	88	2.16	1.43	9.0	10	Most common energy base. Moderately fermentable starch.
☐ Steam-flaked corn	87	2.29	1.57	9.0	9	~ 15 % more energy than dry-rolled. Needs good bunk management (acidosis).

FIGURE 5.1 Block 5. One row per ingredient, with dry matter, energy, protein and fibre.

- 1 The **checkbox** makes it available to the optimiser.
- 2 **Edit** opens its full composition.
- 3 **+ Add** creates a new ingredient (a local by-product, for instance).



WHY ON A DRY BASIS?

Corn silage is 34 % dry matter: two thirds of its weight is water. Comparing a kilo of silage with a kilo of sorghum (90 % DM) would be unfair. On a dry basis everything is compared by what actually feeds; at the end the app gives you back the wet kilos for the wagon.

5.2 Use your lab analyses

If you sent your silage or stover for analysis, press **Edit** and change the values. What weighs most in the result is **dry matter**, **crude protein** and **NDF**; leave the energy if the lab does not report it.

RatioBos Feedlot finishing - West and Bajío ES EN Producer Scientist * ?

Home 1 Animal 2 Region 3 Requirements 4 Management 5 **Ingredients** 6 Prices 7 Formulation 8 Program 9 Dashboard 10 Sustain

Progress: 5 of 11 blocks

Composition: Alfalfa hay (mid-bloom)

1 Table values 2 Close

Name	Category	Dry matter (%) ?	TDN (% MS) ?
Alfalfa hay (mid-bloom)	Dry forages	88	60
NEm (Mcal/kg MS) ?	NEg (Mcal/kg MS) ?	Crude protein (% MS) ?	RDP (% PC) ?
1.32	0.75	18	80
NDF (% MS) ?	Effectiveness factor (0-1) ?	Ether extract (% MS) ?	Calcium (% MS) ?
46	0.8	2.6	1.35
Phosphorus (% MS) ?	Sulfur (% MS) ?	Sodium (% MS) ?	
0.27	0.29	0	

Forage and silage categories count as dietary forage. Fat supplies no energy to rumen microbes.

Ingredient library

New ingredient name

+ Add

Ingredient	MS % ?	ENm ?	ENg ?	CP % ?	NDF % ?	Notes	
DRY FORAGES							
☒ Alfalfa hay (mid-bloom)	88	1.32	0.75	18.0	46	Ca, P and S from fresh alfalfa in the same table.	Edit
☐ Oat hay	91	1.17	0.62	9.0	59	Minerals from fresh oat forage.	Edit
☐ Sudangrass hay	91	1.19	0.57	9.0	67	Boot stage.	Edit

FIGURE 5.2 An ingredient's composition. Every field has its ?.

1

↶ **Table values** undoes your changes.

2

Close



IN THE PEN

We have 12 ingredients ticked: alfalfa hay, corn stover, corn silage, dry-rolled corn, dry-rolled sorghum, molasses, DDGS, 44 % soybean meal, urea, limestone, salt and the trace-mineral and vitamin premix.

6

Local prices and inclusion limits



The price you enter here is what the app minimises. A stale price gives a ration that is no longer the cheapest.

6.1 Enter what you pay

Enter the price per kilo **as you buy it and delivered to the ranch** (freight and unloading included). The app converts it to price per kilo of dry matter, which is what it compares.

Ingredient	\$/kg AF ?	\$/kg MS ?	Minimum % ?	Maximum %	\$ per Mcal NEg ?	\$ per kg CP ?
Alfalfa hay (mid-bloom)	5.22	5.93	0		7.91	32.95
Corn stover	1.66	1.95	0	50	4.65	32.55
Corn silage	1.23	3.62	0		3.58	45.22
Dry-rolled corn	5.03	5.72	0		4.00	63.51
Dry-rolled sorghum	4.23	4.70	0		3.38	42.73
Cane molasses	3.34	4.58	0	10	4.24	50.84
Distillers grains with solubles (DDGS)	5.4	6.07	0	30	3.99	19.57
Soybean meal 44 %	8.6	9.66	0		7.21	19.72
Urea (46 % N)	9.5	9.60	0	1	—	3.33
Calcium carbonate	1.6	1.62	0	2	—	—
Salt (NaCl)	3.2	3.23	0	0.6	—	—
Trace-mineral and vitamin premix	28	29.47	0.3	0.3	—	—

AF = as fed (wet). To fix an exact inclusion (e.g. premix 0.3 %) enter the same value as minimum and maximum.

FIGURE 6.1 Block 6. Prices and limits for the 12 ticked ingredients.

- 1 **\$/kg AF:** your price, as fed.
- 2 **Minimum %** of dry matter: forces at least that much in.
- 3 **Maximum %:** the cap for animal health, palatability or logistics.

6.2 Compare prices with two columns

The \$ per Mcal NEg and \$ per kg CP columns tell you what each unit of energy and protein costs in each ingredient. That is the right way to compare: the sorghum in our case gives the cheapest energy (\$3.38 per Mcal) and DDGS the cheapest protein after urea (\$19.57 per kg CP).



DECISION RULE · LIMITS WORTH KEEPING

Urea	At most 1 % of DM, always well mixed. More risks toxicity.
DDGS	At most 30 %: they bring sulfur and fat.
Molasses	At most 10 % in grain-based finishing.
Whole cottonseed	At most 15 % because of gossypol.
Limestone	At most 2 %: it is a calcium source, not a filler.
Premix	Minimum = maximum = the maker's dose (0.3 % in our case).



TIP · TO FIX AN EXACT INCLUSION

Type the same value as minimum and maximum. Use it for the premix or for an ingredient you have already bought and want to use up.



7

Least-cost formulation

One button. The app searches every possible blend for the cheapest one that covers what your cattle need, and tells you how much they will gain and what each kilo costs.

7.1 Press Formulate

RatioBos Feedlot finishing · West and Bajío ES EN **Producer** Scientist * ?

gion 3 Requirements 4 Management 5 Ingredients 6 Prices 7 **Formulation** 8 Program 9 Dashboard 10 Sustainability 11 Report ✓ Valid

Progress: 7 of 11 blocks

IN A NUTSHELL
Press Formulate: the app finds the cheapest ration that covers your cattle's needs and gives you the mixer sheet.

Feedlot finishing Zebu × European cross 380 kg · 1.35 kg/d 12 ingredients

1 **Formulate**

No result yet (or you changed an input). Press **Formulate** and the scale will find the balance between nutrients and cost.

Evaluate an existing formula ?
Type the dry-basis percentages of a ration you already use (or from a paper). The app computes its supply, its cost and the gain it allows.

Alfalfa hay (mid-bloom)	Corn stover	Corn silage	Dry-rolled corn
0	0	0	0
Dry-rolled sorghum	Cane molasses	Distillers grains with solubles (DDGS)	Soybean meal 44 %
0	0	0	0
Urea (46 % N)	Calcium carbonate	Salt (NaCl)	Trace-mineral and vitamin premix
0	0	0	0

Evaluate formula Copy the optimal formula here Clear

FIGURE 7.1 Block 7. At the top, a summary of what you are about to formulate.

1 **Formulate**. It takes under a second.

? WHAT DOES THE APP DO IN THAT SECOND?

It solves a **linear programming** problem: among every combination of your ingredients that meets each minimum and maximum, it finds the one with the lowest cost per kilo of dry matter. Then it checks whether that ration's energy matches the one it assumed to estimate intake; if not, it corrects and solves again until they match. Finally it proves with a mathematical certificate that no cheaper one exists.

7.2 The four result cards

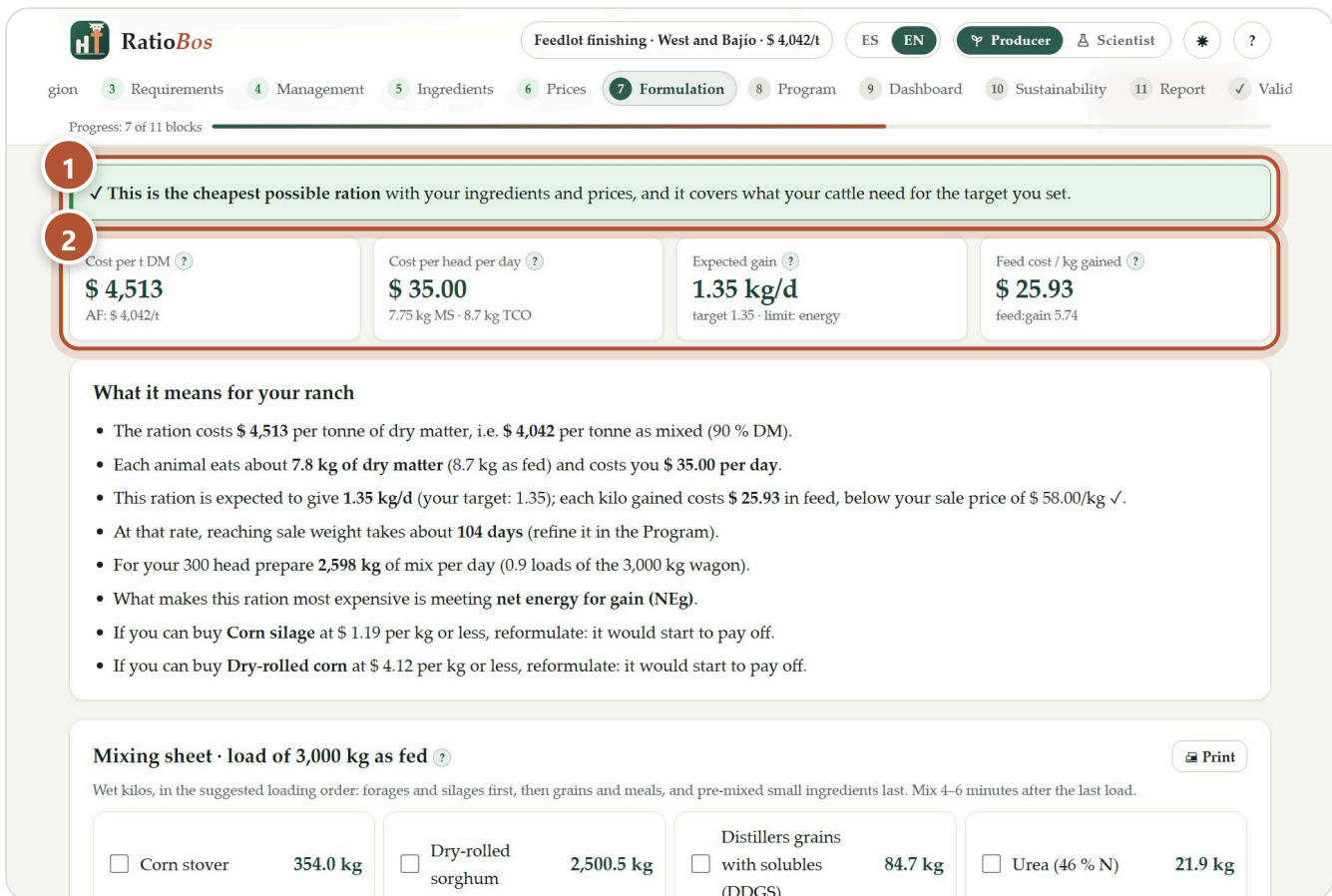
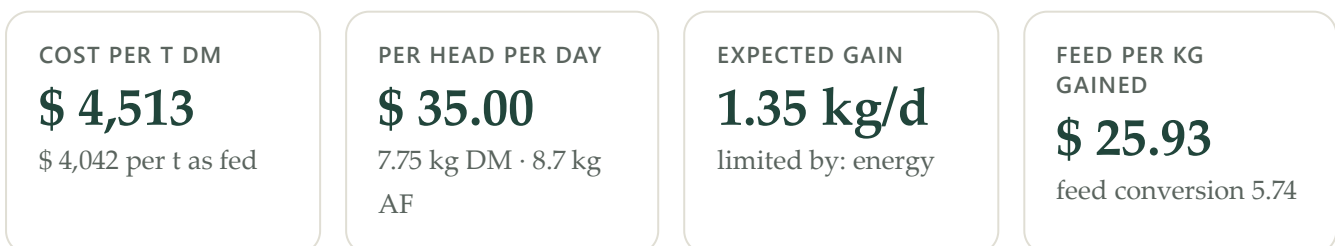


FIGURE 7.2 Result. The green message confirms it is the cheapest possible ration.

1 Least-cost confirmation.

2 The **four cards** explained below.



Cost per t DM What a tonne of the ration costs without water. Below, the same tonne wet, as it leaves the wagon.

Cost per head per day Expected intake $\times$ price. Each head's daily feed bill.

Expected gain The gain the ration allows, and what limits it: energy or protein.

Feed cost per kg gained The business number. It must sit well below your sale price.

7.3 What it means for your ranch

Below the cards, the app turns the numbers into sentences: how much each animal eats, how many days remain to sale weight, how many kilos to mix per day and which ingredient would start to pay off if its price fell.

RatioBos Feedlot finishing · West and Bajio · \$ 4,042/t ES EN **Producer** Scientist * ?

gion 3 Requirements 4 Management 5 Ingredients 6 Prices 7 **Formulation** 8 Program 9 Dashboard 10 Sustainability 11 Report ✓ Valid

Progress: 7 of 11 blocks

1.274 kg MS · 8.7 kg TCO target 1.35 · limit: energy need: gain 3.7/4

What it means for your ranch

- The ration costs \$ 4,513 per tonne of dry matter, i.e. \$ 4,042 per tonne as mixed (90 % DM).
- Each animal eats about **7.8 kg of dry matter** (8.7 kg as fed) and costs you \$ **35.00 per day**.
- This ration is expected to give **1.35 kg/d** (your target: 1.35); each kilo gained costs \$ **25.93** in feed, below your sale price of \$ 58.00/kg ✓.
- At that rate, reaching sale weight takes about **104 days** (refine it in the Program).
- For your 300 head prepare **2,598 kg** of mix per day (0.9 loads of the 3,000 kg wagon).
- What makes this ration most expensive is meeting **net energy for gain (NEg)**.
- If you can buy **Corn silage** at \$ 1.19 per kg or less, reformulate: it would start to pay off.
- If you can buy **Dry-rolled corn** at \$ 4.12 per kg or less, reformulate: it would start to pay off.

Mixing sheet · load of 3,000 kg as fed ?

Wet kilos, in the suggested loading order: forages and silages first, then grains and meals, and pre-mixed small ingredients last. Mix 4–6 minutes after the last load. [Print](#)

☐ Corn stover 354.0 kg	☐ Dry-rolled sorghum 2,500.5 kg	☐ Distillers grains with solubles (DDGS) 84.7 kg	☐ Urea (46 % N) 21.9 kg
☐ Calcium carbonate 25.0 kg	☐ Salt (NaCl) 5.52 kg	☐ Trace-mineral and vitamin premix 8.48 kg	

Total: 3,000 kg · load cost: \$ 12,125.09 · feeds 346 head-days

FIGURE 7.3 What it means for your ranch. Always in plain language.



IN THE PEN

The ration costs **\$4,042 per tonne** as mixed. Each bull eats 8.7 kg a day and costs **\$35.00**. Expected gain is **1.35 kg/d**, at **\$25.93 of feed per kilo gained**: less than half the \$58 we will sell it for. For the 300 animals we must prepare **2,598 kg a day**, just under one wagon load, and they will reach sale weight in about 104 days.

7.4 The mixer-wagon sheet

This is what goes to the pen. The kilos are **wet**, as the wagon scale weighs them, and come in **loading order**: forages and silages first, then grains and meals, and the small pre-blended ingredients last. Each line has a box to tick as you load.

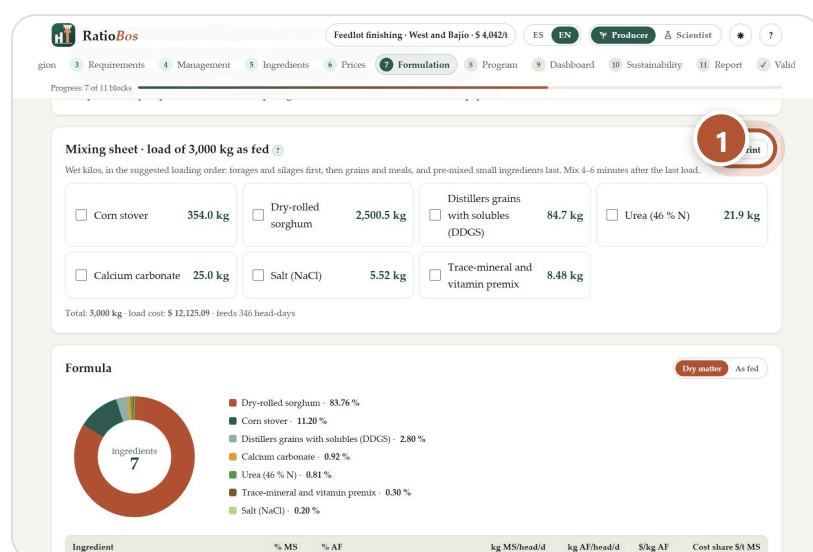


FIGURE 7.4 Mixing sheet for a 3,000 kg load.

1 **Print** prints just the sheet, on one page.

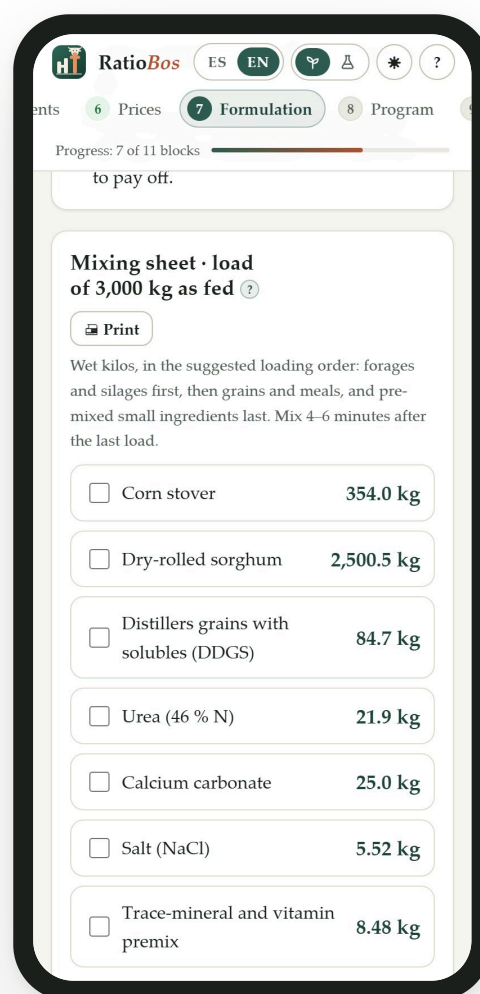


FIGURE 7.5 On a phone, by the wagon.

ORDER	INGREDIENT	KG
1	Corn stover	354.0
2	Dry-rolled sorghum	2,500.5
3	Distillers grains (DDGS)	84.7
4	Urea · limestone · salt · premix (<i>pre-blended</i>)	21.9 · 25.0 · 5.52 · 8.48
Load total · \$12,125.09 · feeds 346 head-days		3,000

! CAUTION · UREA, SALT AND PREMIX

Together they are 61 kg in a 3,000 kg load. Tipped straight into the wagon they spread badly and some animals get too much. **Blend them first with about 50 kg of ground sorghum** and add that premix last. Poorly mixed urea can poison cattle.

7.5 The formula and the rumen traffic light

The formula appears as a doughnut and a table, on a dry-matter or as-fed basis (button at the top right). The table also shows how many kilos of each ingredient each animal eats and how much each adds to the cost.

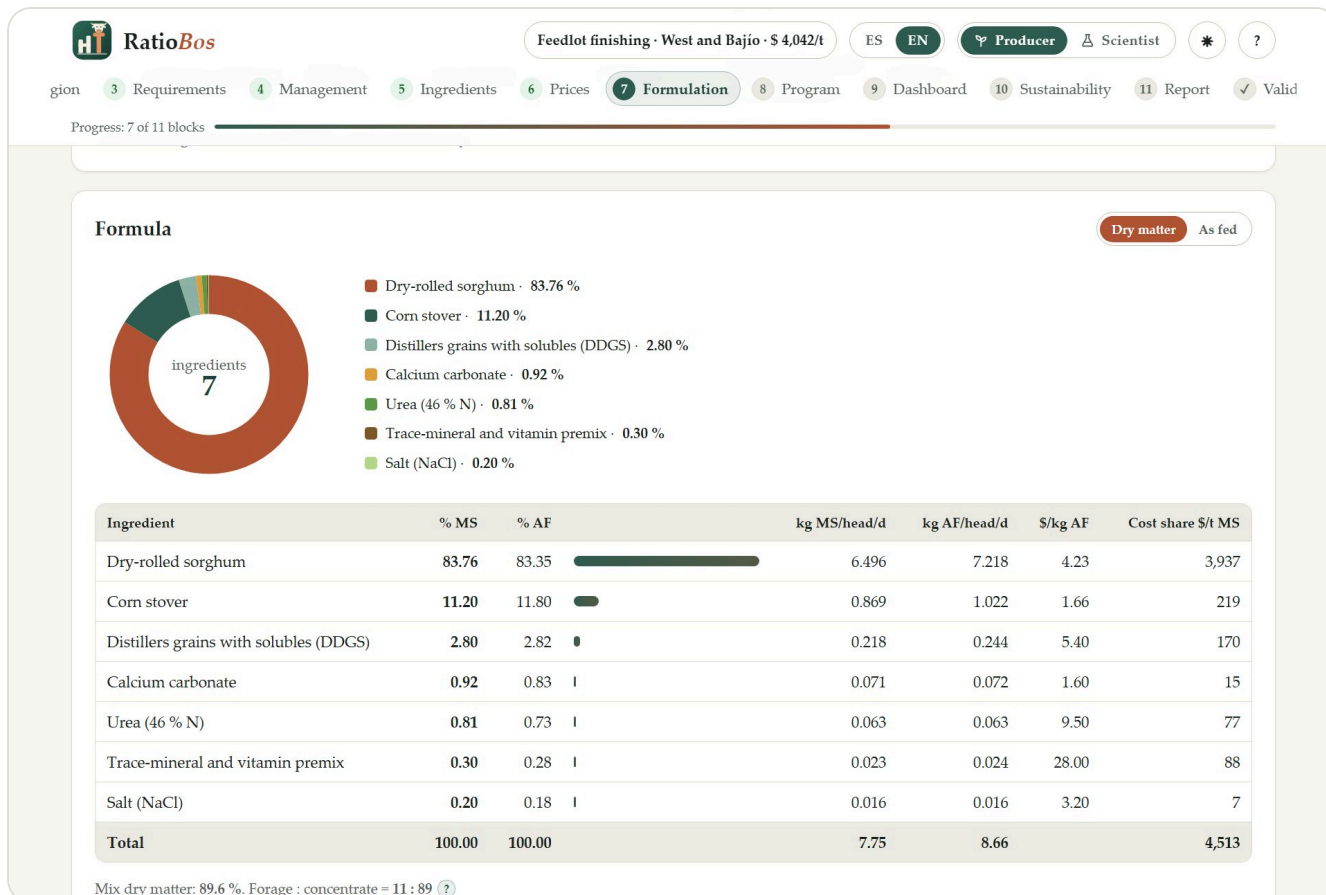


FIGURE 7.6 Formula on a dry-matter basis. Sorghum 83.76 %, stover 11.20 %, DDGS 2.80 %, limestone 0.92 %, urea 0.81 %, premix 0.30 % and salt 0.20 %.

Further down, two bars compare the day's energy and protein against what the animal needs, and the **rumen-health traffic light** checks what least cost might neglect.

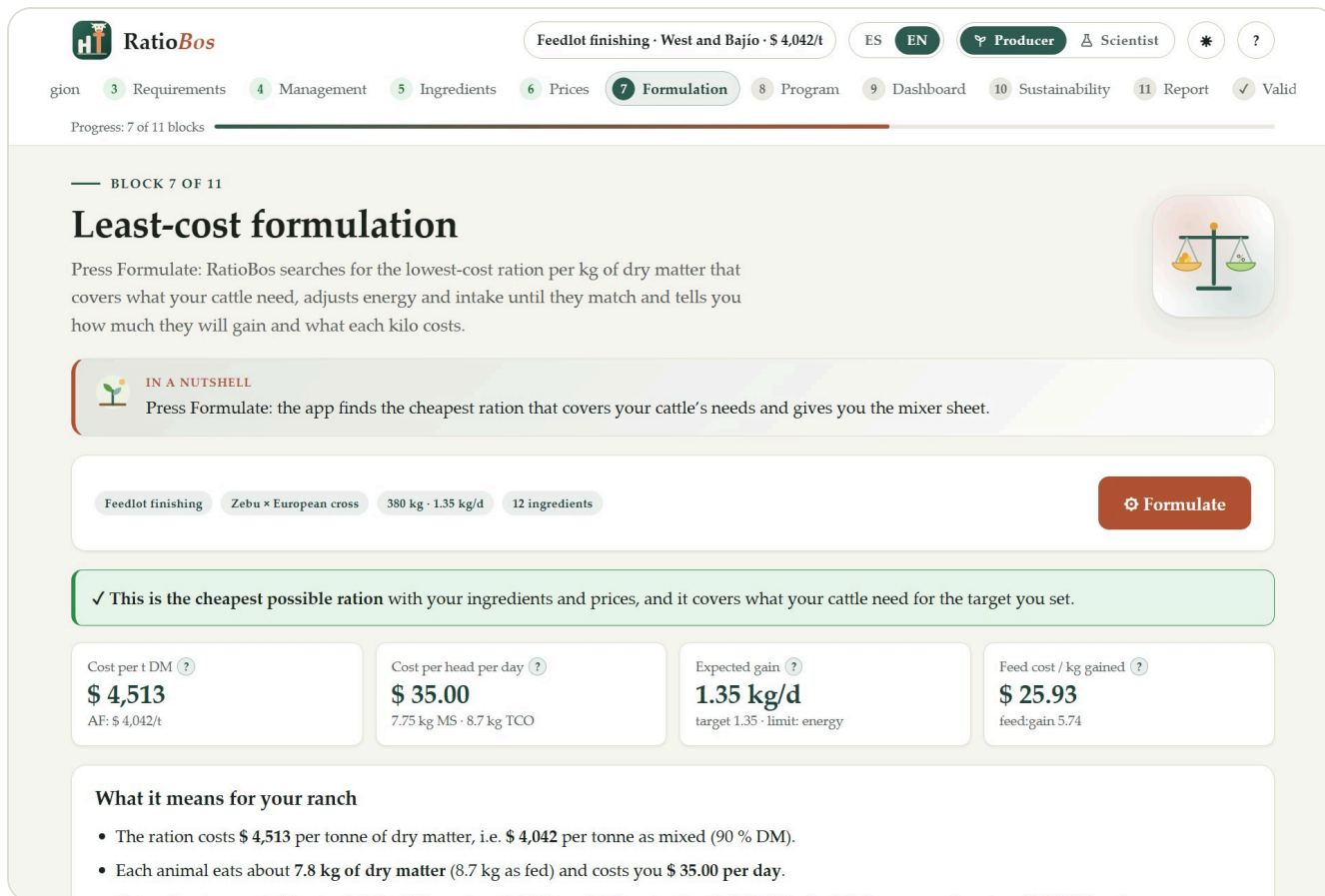


FIGURE 7.7 The day's energy and protein, rumen health and footprint.



DECISION RULE · RUMEN TRAFFIC LIGHT

Effective fibre (peNDF)	Below 7 % in finishing: risk of acidosis and bloat. Add stover.
Total fat	Above 6–7 % slows fibre digestion.
Urea	Amber above 1 % of DM or 25 % of the protein; red above 1.2 % or 35 %: toxicity risk.
Sulfur	Amber above 0.30 %, red above 0.40 %: polioencephalomalacia. The light does not count water; if your water is high in sulfate, stay well below the limit.
Ca : P	Green from 1.5 : 1; amber between 1.2 and 1.5; red below 1.2. A low Ca : P raises the risk of urinary stones in males: add a little limestone.



IN THE PEN

All green except **Ca : P**, in amber (1.49 : 1, just under 1.5). In bulls it is worth fixing: if in Block 6 we raise the limestone minimum to **1.1** % and formulate again, Ca : P rises to **1.69 : 1** and a tonne of DM costs only \$2 more (\$4,515). To keep the example's numbers, the following chapters stay with the original formula.

7.6 Which ingredients are worth watching?

The **Unused ingredients · entry price** table tells you at what price an ingredient left out would start to pay off. Use it when negotiating with your supplier.

Feedlot finishing · West and Bajío · \$ 4,042/t
ES EN
🔧 Producer
🔬 Scientist
⚙️
?

gion
3 Requirements
4 Management
5 Ingredients
6 Prices
7 Formulation
8 Program
9 Dashboard
10 Sustainability
11 Report
✓ Valid

Progress: 7 of 11 blocks

Which ingredients are worth watching?

Unused ingredients · entry price ?

Ingredient	Current price	Would enter at
Alfalfa hay (mid-bloom)	5.22	≤ 2.86
Corn silage	1.23	≤ 1.19
Dry-rolled corn	5.03	≤ 4.12
Cane molasses	3.34	≤ 2.64
Soybean meal 44 %	8.60	≤ 5.58

Prices per kg as fed.

Evaluate an existing formula ?

Type the dry-basis percentages of a ration you already use (or from a paper). The app computes its supply, its cost and the gain it allows.

Alfalfa hay (mid-bloom)	Corn stover	Corn silage	Dry-rolled corn
0	0	0	0
Dry-rolled sorghum	Cane molasses	Distillers grains with solubles (DDGS)	Soybean meal 44 %
0	0	0	0
Urea (46 % N)	Calcium carbonate	Salt (NaCl)	Trace-mineral and vitamin premix
0	0	0	0

FIGURE 7.8 Entry price. If corn silage drops from \$1.23 to \$1.19, it enters the ration.

7.7 If “No solution” appears

It means no blend of your ingredients meets everything at once. In growing animals, the usual cause is a gain target higher than the ingredients or the climate allow. The app works out the **maximum feasible gain** and offers a button to use it.

Feedlot finishing · Gulf (humid tropics)
ES EN
🔧 Producer
🔬 Scientist
⚙️
?

gion
3 Requirements
4 Management
5 Ingredients
6 Prices
7 Formulation
8 Program
9 Dashboard
10 Sustainability
11 Report
✓ Valid

Progress: 7 of 11 blocks

No solution.

No combination of the selected ingredients meets all the constraints.

With these ingredients and limits, the maximum feasible gain is 1.30 kg/d. **1 Use 1.30 kg/d and formulate** To gain more, add a higher-energy ingredient (steam-flaked corn, fat).

Relaxing a single constraint is not enough: lower the gain target or add ingredients (a grain for energy, a protein meal, urea, limestone).

Evaluate an existing formula ?

Type the dry-basis percentages of a ration you already use (or from a paper). The app computes its supply, its cost and the gain it allows.

Alfalfa hay (mid-bloom)	Corn stover	Corn silage	Dry-rolled corn
0	0	0	0
Dry-rolled sorghum	Cane molasses	Distillers grains with solubles (DDGS)	Soybean meal 44 %
0	0	0	0
Urea (46 % N)	Calcium carbonate	Salt (NaCl)	Trace-mineral and vitamin premix
0	0	0	0

Evaluate formula
Copy the optimal formula here
Clear

← Prices
Next
8 · Feeding program →

FIGURE 7.9 The same pen in Veracruz, dry season. In that heat the 1.35 kg/d target is not possible: the maximum is 1.30.

1 Use 1.30 kg/d and formulate .

What do I do if there is no solution? Try in this order:

1

Does the app offer a maximum gain?

Use it, or add a higher-energy ingredient (steam-flaked corn, fat) to reach your target.

2

Did you edit the Block 3 table?

Press *↺ Recompute from the animal* and formulate again.

3

Does it say “No selected ingredient supplies...”?

Tick one in Block 5 that supplies it: limestone for calcium, salt for sodium, a meal for protein.

4

Still no solution?

The app lists the constraints where relaxing **just one** would give a solution. Check those limits in Block 6.



8

The whole stay in the pen, day by day: how many days it takes, how much feed and money it uses and what is left per head.

8.1 The phases

For a feedlot the app proposes three phases: **receiving and step-up** (plenty of forage so the rumen adapts to grain), **growing** and **finishing**. For each you can change its end weight, gain target and forage range. Enter the **sale weight** and press **Formulate the program**.

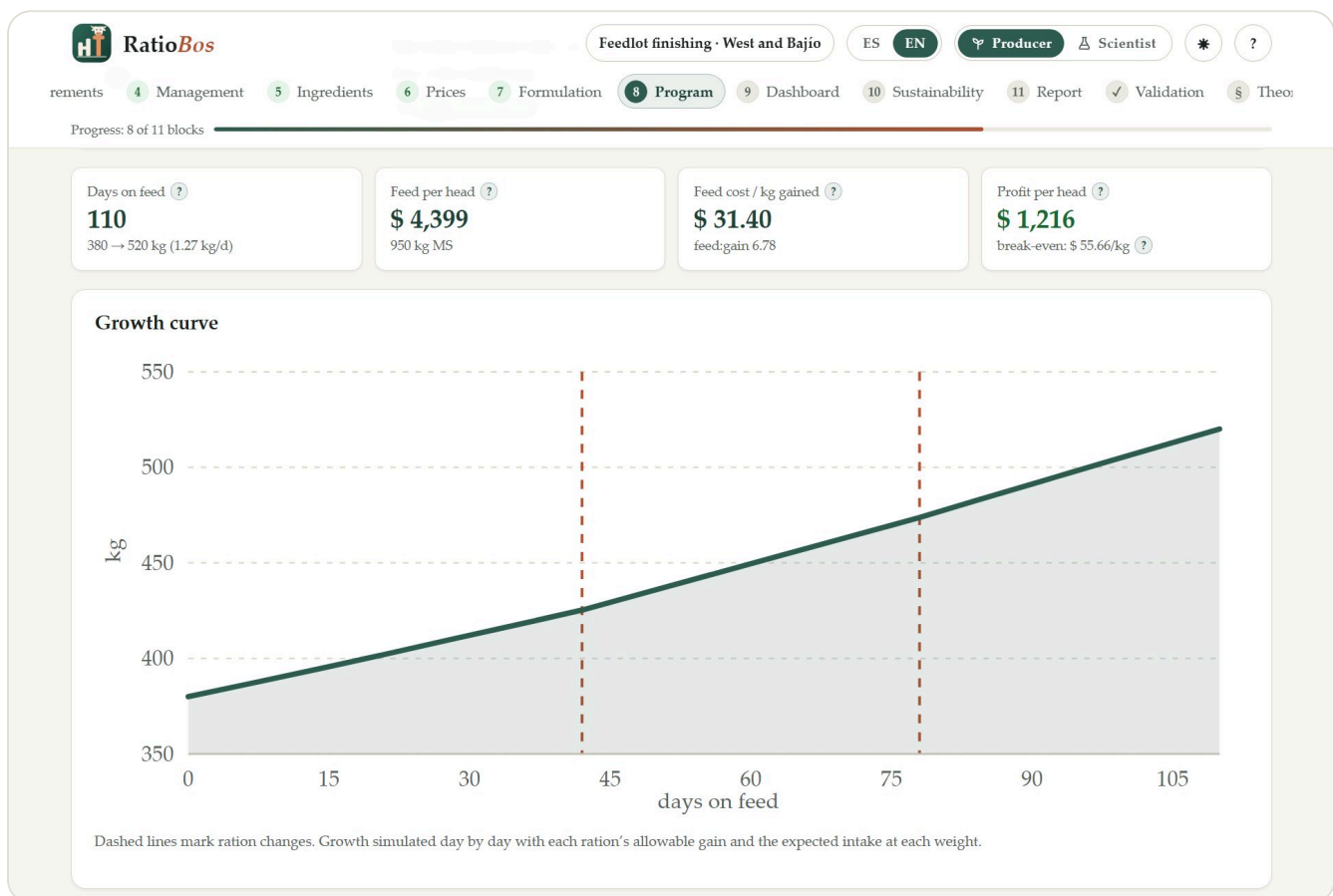


FIGURE 8.1 Block 8. Four cards sum up the whole feeding period.

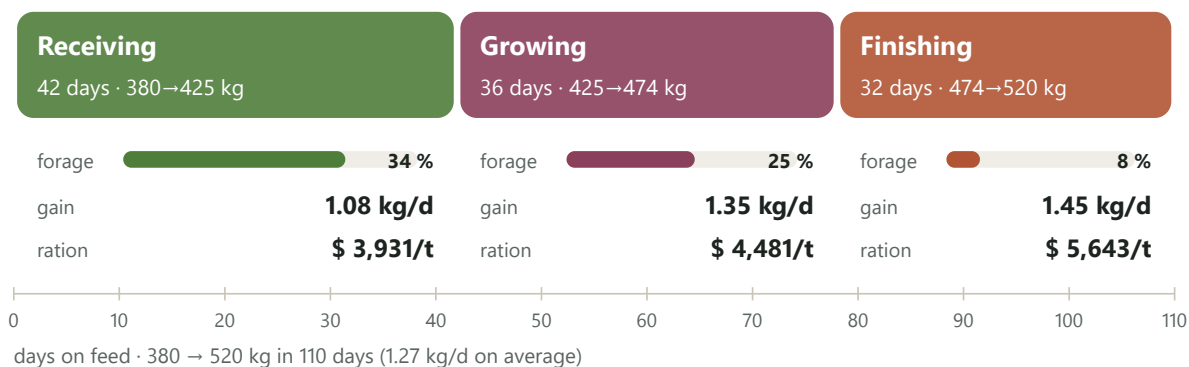


FIGURE 8.2 The program at a glance. As forage drops, gain rises, but so does the ration cost. Data from the case program.

? WHY IS THE PROGRAM'S GAIN LOWER THAN THE TARGET?

Because receiving, with 34 % forage, gains less (1.08 kg/d) and because in every phase the app grows the animal at the gain the ration **really** allows, day by day. The 110-day average is 1.27 kg/d. For the same reason the program's cost per kilo (\$31.40) is higher than that of the Block 7 ration (\$25.93), computed for the 380 kg bull: finishing is dearer and the heavier animal converts worse.

8.2 The growth curve

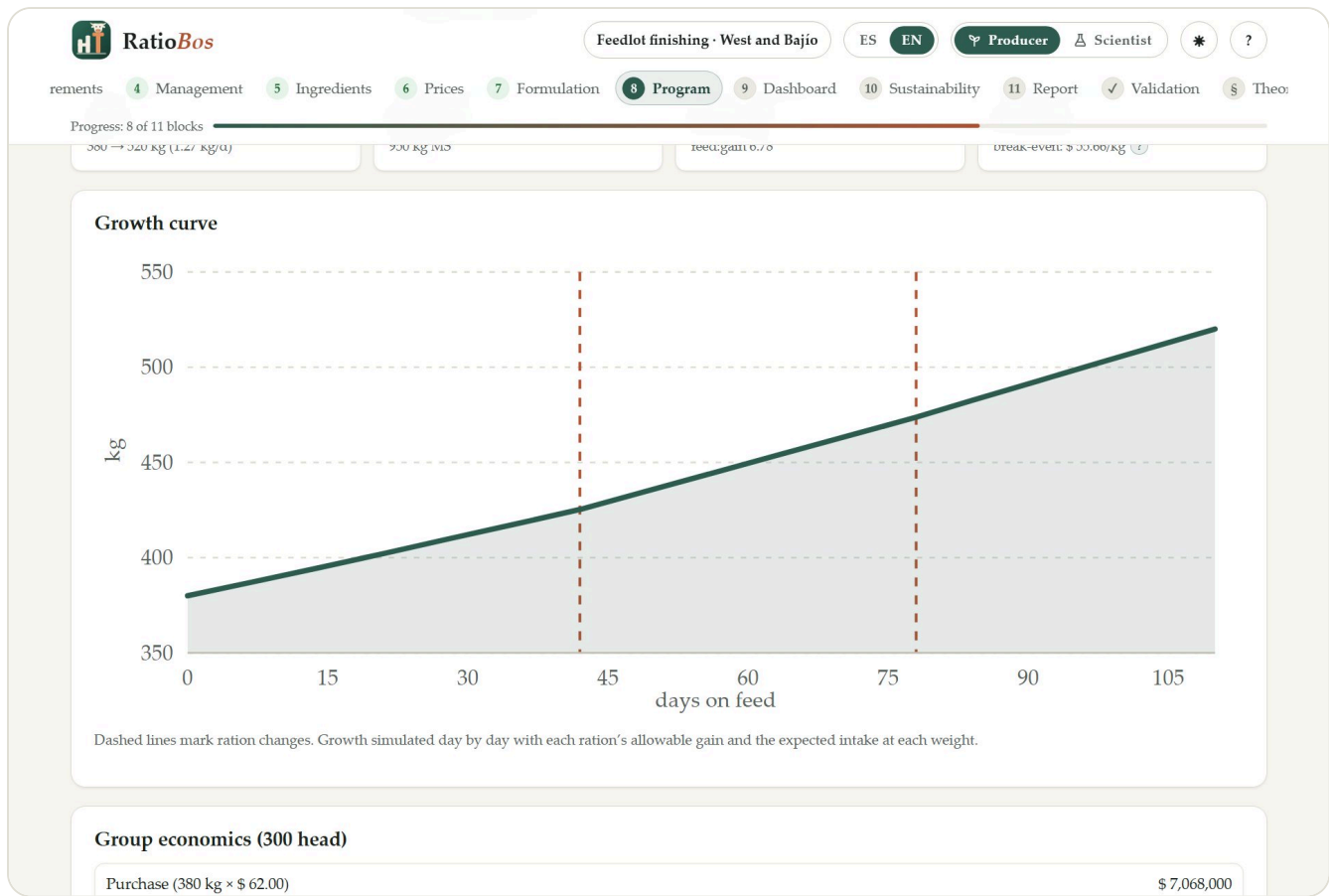


FIGURE 8.3 Growth curve. Dashed lines mark the ration changes.

8.3 Batch economics

RatioBos Feedlot finishing - West and Bajio ES EN Producer Scientist * ?

Management 4 Ingredients 5 Prices 6 Formulation 7 **Program 8** Dashboard 9 Sustainability 10 Report 11 Validation ✓ Theoretical \$

Progress: 8 of 11 blocks

BLOCK 8 OF 11

Feeding program

Simulates the whole stay in the pen: RatioBos formulates a ration for each phase and grows the animal day by day at the gain that ration allows. You get days on feed, total feed, cost per kg gained and profit per head.

IN A NUTSHELL
Simulate the whole feeding period: how many days, how much feed and money it takes and what you keep per head.

Feedlot finishing - Zebu x European cross - Bull Entry: 380 kg **Formulate the program**

Phase	Up to (kg)	Gain target (kg/d)	Forage min. %	Forage max. %
Receiving and step-up	425	1.1	30	45
Growing	473	1.35	15	25
Finishing	520	1.45	8	12

Sale weight (kg) ? 520 Other limits (effective fiber, fat, sulfur) are those of the animal type. [Suggested phases](#)

Days on feed ? Feed per head ? Feed cost / kg gained ? Profit per head ?

FIGURE 8.4 Batch economics and phase summary.

1 **Download program (CSV)** to open it in Excel.



IN THE PEN

The program takes **110 days** from 380 to 520 kg. Each bull eats 950 kg of dry matter, costing **\$4,399**. Profit is **\$1,216 per head** and **\$364,943** for the lot. The **break-even price** is \$55.66: if live weight sells below that, the lot loses money. With a margin of only \$2.34 per kilo over break-even, it pays to agree the sale before buying.

Results dashboard



9

Every number that matters in one view and in charts: what the ration achieves, the most profitable gain, the cost in each region and the grain-price risk.

9.1 Achieved performance

The table sets the **current ration** beside the **whole program** with the four classic feedlot indicators.

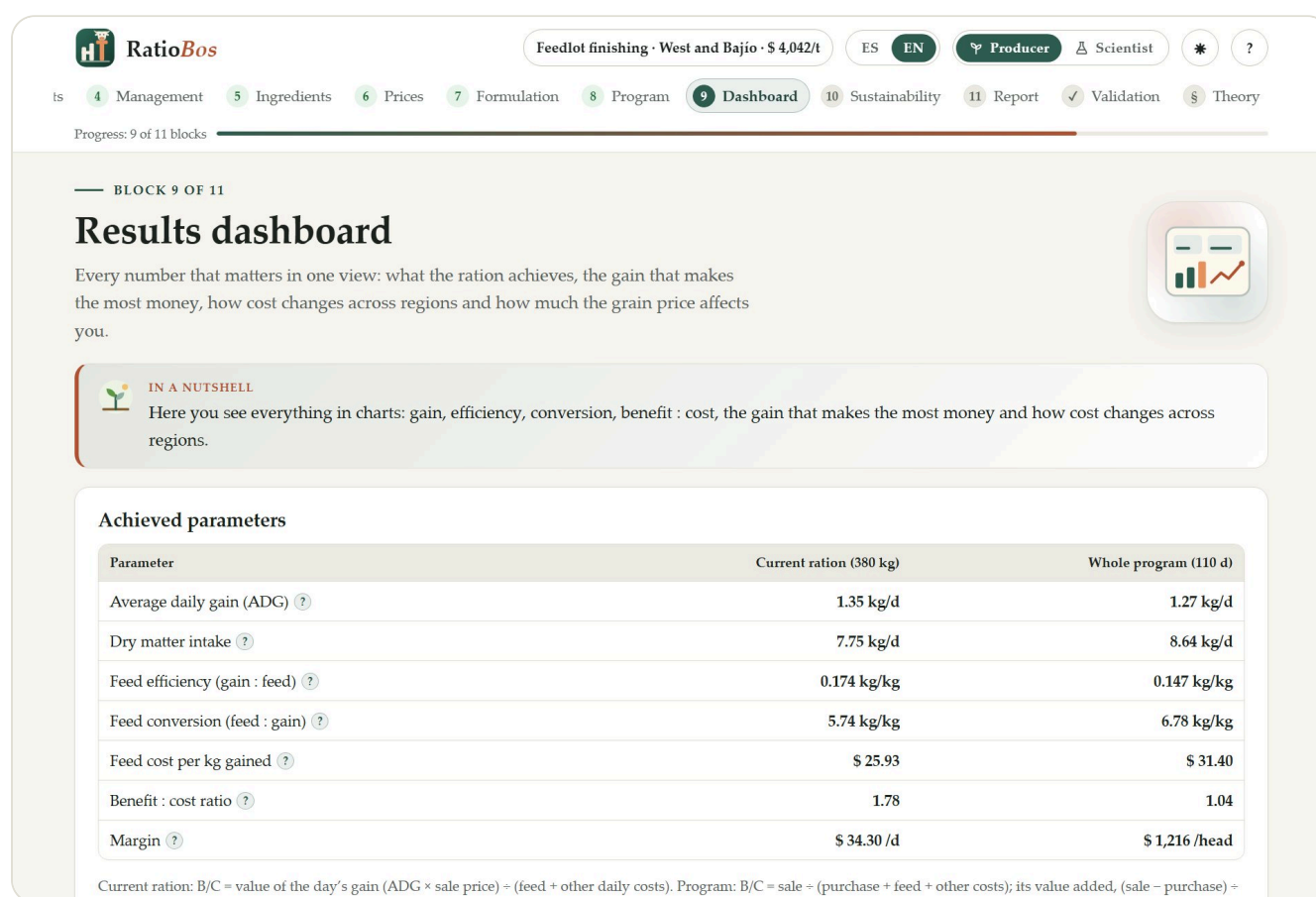


FIGURE 9.1 Achieved performance. Each indicator has its ? with the decision scale.

INDICATOR	HOW IT IS COMPUTED	RATION	PROGRAM
Average daily gain (ADG)	kg gained ÷ days	1.35	1.27
Feed efficiency	gain ÷ feed (higher is better)	0.174	0.147
Feed conversion	feed ÷ gain (lower is better)	5.74	6.78
Benefit : cost	money in ÷ money out (over 1 = profit)	1.78	1.04



WHY TWO SUCH DIFFERENT BENEFIT : COST RATIOS?

The ration ratio compares the value of what the animal gains in a day with what feeding it costs that day: 1.78 says each peso of feed and management produces \$1.78 of beef. The program ratio also includes **buying the animal**, the biggest investment: 1.04 is the real return on the whole business. The app also shows the **value added** (sale minus purchase, over feed and other costs): 1.23.

9.2 Which gain pays best?

Faster gain does not always pay: each extra kilo demands a dearer ration. The app formulates for several gain targets and plots two curves: the **margin per head per day** (green) and the **feed cost per kilo gained** (terracotta). The circle marks the economic optimum.

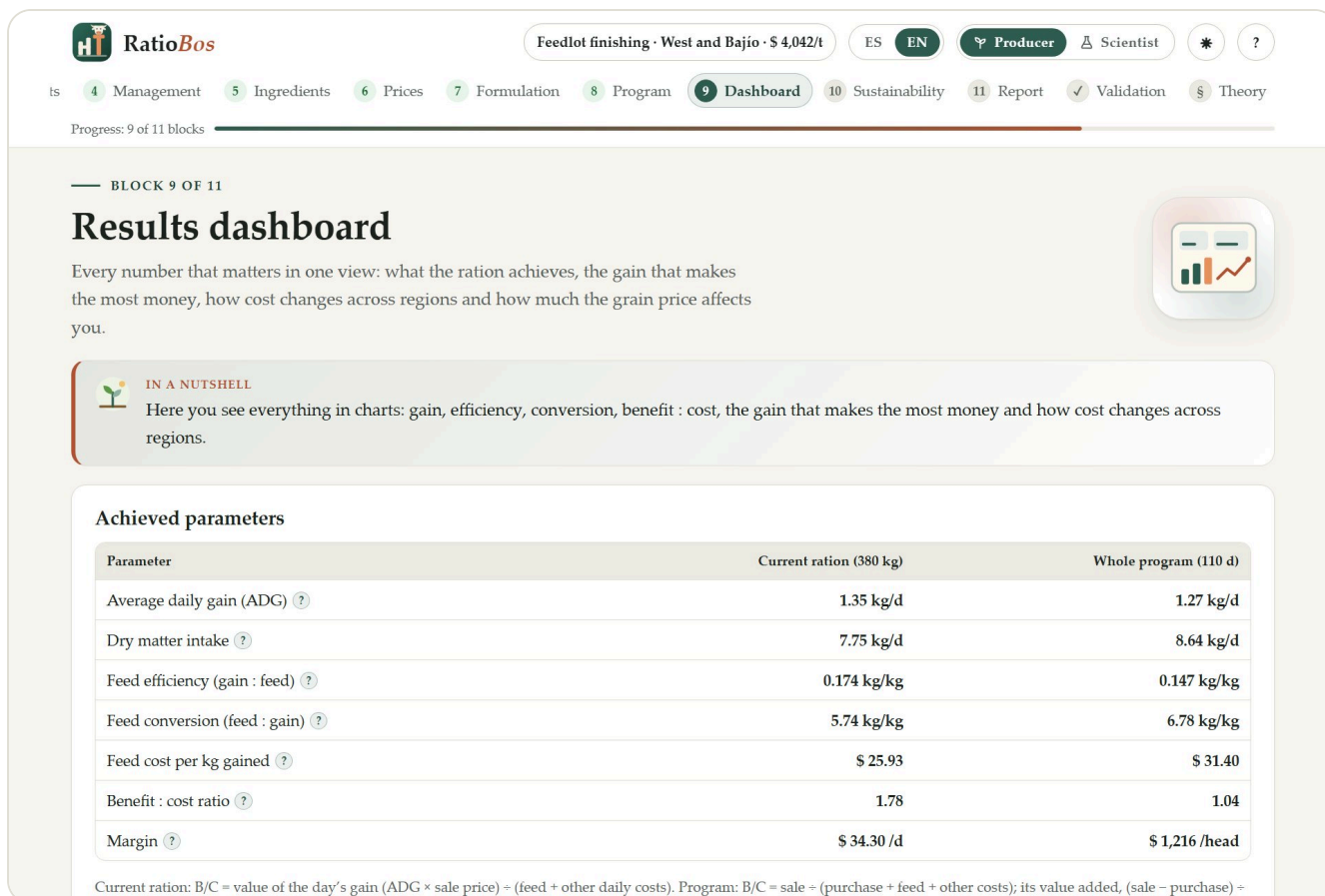


FIGURE 9.2 The optimal gain. The dashed line is your ration.



IN THE PEN

At our prices, the highest daily profit comes at **1.40 kg/d**: \$36.44 per head per day against \$34.30 at our 1.35 target. The gap is small: the target is already close to the optimum and the app says so with a tick.

9.3 The same animal in every region

The app repeats your formulation in all 9 regions, with the season's climate, your shade and each region's estimated prices. If a region's climate does not allow your target, it uses the maximum feasible gain there and marks it with an asterisk.

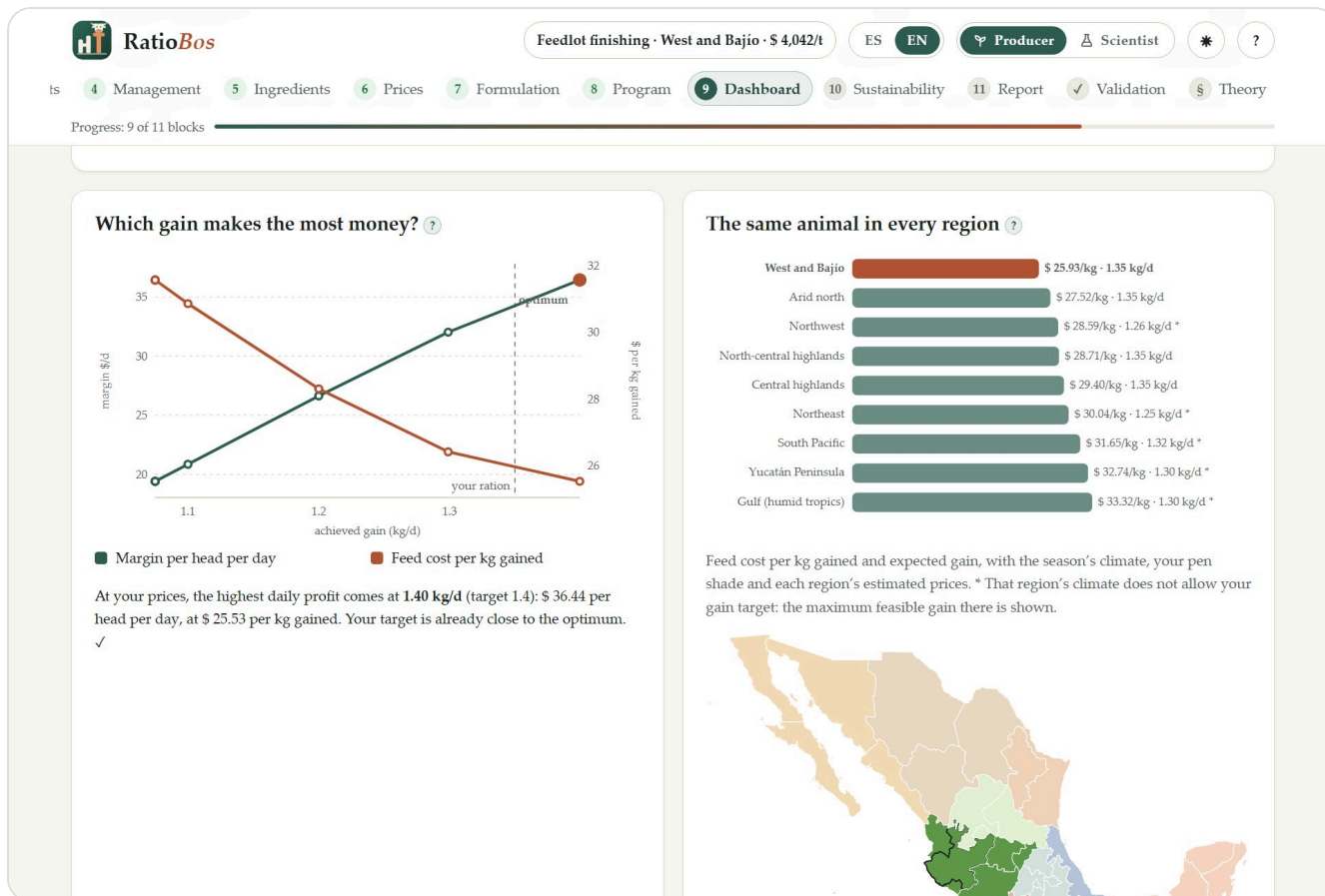


FIGURE 9.3 Feed cost per kilo gained in each region, dry season. Your region is in terracotta.

9.4 If the grain price changes

The chart moves the price of the ingredient that weighs most in the cost (sorghum, in our case) from 70 % to 130 % of today's and reformulates at each point. The green line is the cost per tonne; the terracotta one, how much sorghum stays in the ration.

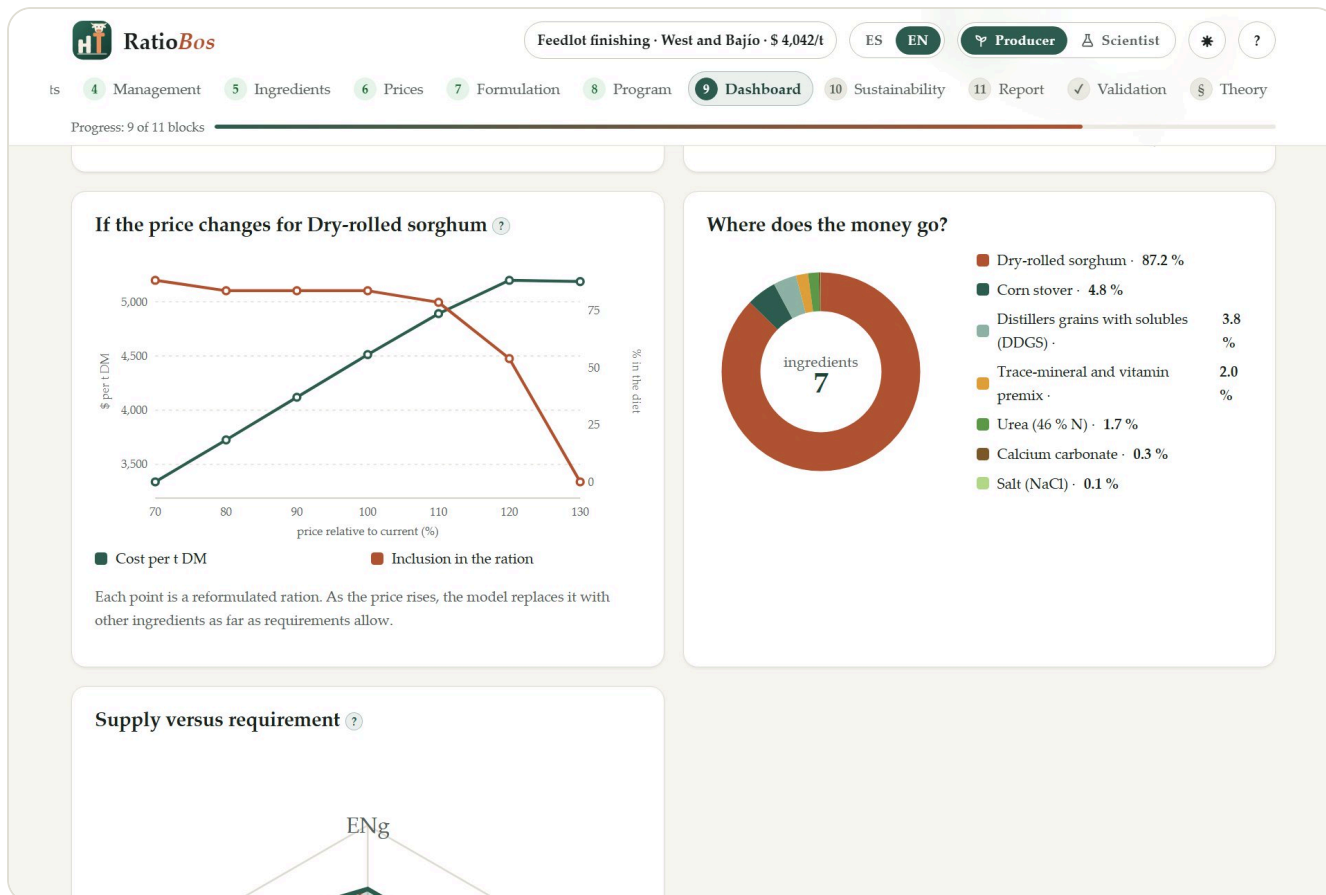


FIGURE 9.4 Sorghum-price sensitivity and where the money goes.



TIP · HOW TO READ IT

While the terracotta line is flat, sorghum is irreplaceable and its price hits your cost in full: that is the ingredient to buy ahead or under contract. When the line drops, the app has found a substitute. In our case, from a 10 % rise it starts replacing it with silage, molasses and DDGS, and from 25 % it switches entirely to dry-rolled corn.

9.5 Compare yourself with published feedlots

At the end of the dashboard is a table with gain, intake, efficiency, conversion and benefit : cost from experimental feedlots published in Mexico and the southern United States, from CIPES in Sonora to Sinaloa, Zacatecas and El Centro, California. Use it to check that your numbers are reasonable.

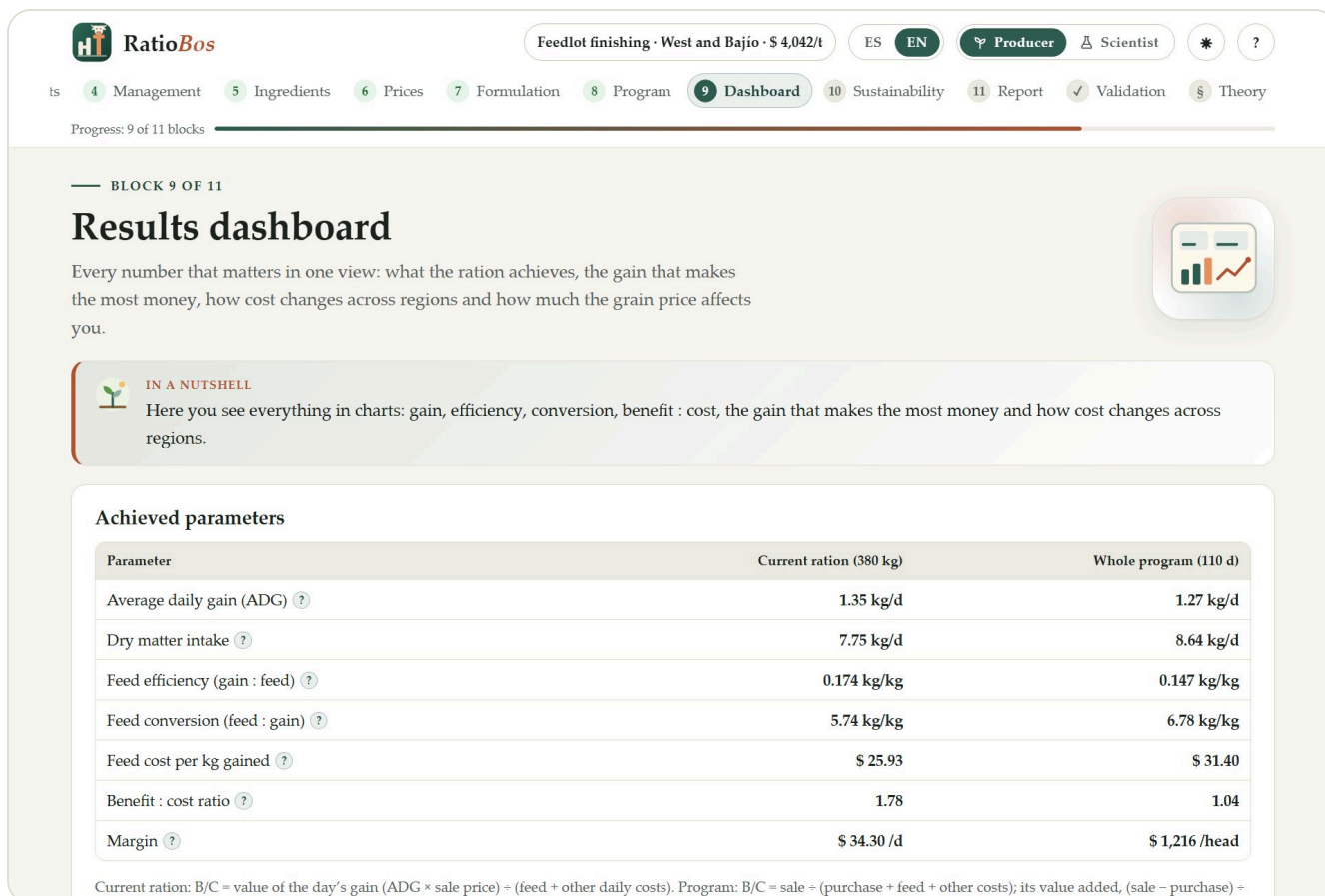


FIGURE 9.5 Published performance from reference feedlots.

Environmental footprint and sustainable production



10

Producing beef uses water and land and leaves gases, nutrients and waste. Here you see how much, and look for the balance between profit and the environment.

RatioBos uses the word **sustainability** as FAO defines it: producing today without compromising the capacity to produce tomorrow, in its three dimensions at once: **environmental**, **economic** and **social**. That is why this block does not seek the “greenest” ration at any price, but the one that cuts the footprint and still pays.

10.1 The sustainability traffic light

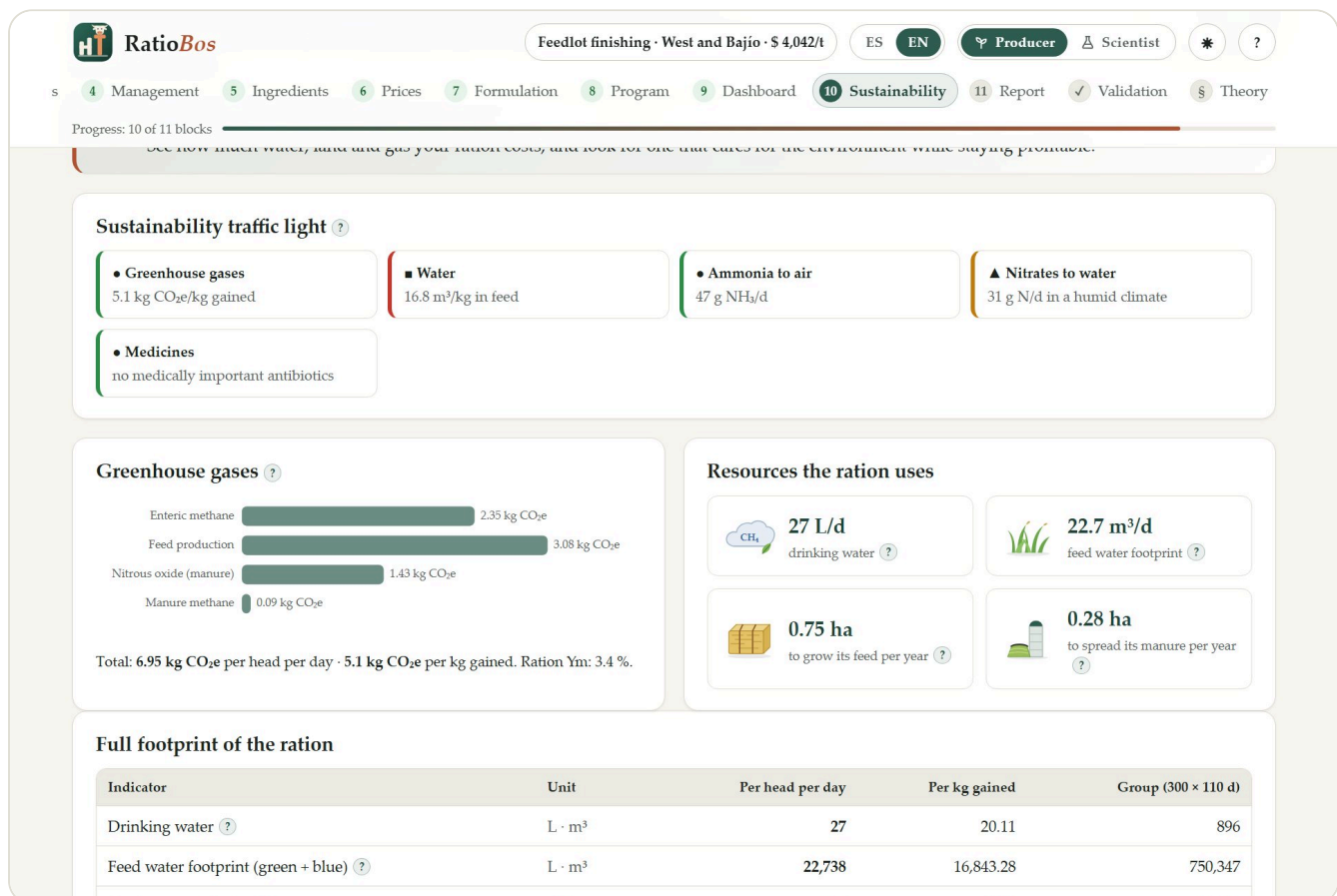


FIGURE 10.1 Sustainability traffic light, gases and resources.

LIGHT	WHAT IT MEASURES	IN OUR CASE
Greenhouse gases	Methane from the rumen and manure, nitrous oxide and feed production, in CO ₂ equivalent per kg gained.	5.1 kg
Water	Rain and irrigation water it took to grow the feed, per kg gained.	16.8 m ³
Ammonia to air	Nitrogen volatilised from manure: odour, fine dust and acid rain.	47 g/d
Nitrates to water	Nitrogen that can leach to wells and rivers in a wet climate.	31 g/d
Medicines	Whether you use antibiotics important in human medicine (WHO list).	none

10.2 Gases and resources

The gas bars split where each kilo of CO₂ equivalent comes from. In our case **feed production** (3.08 kg a day) outweighs **rumen methane** (2.35 kg): on a high-grain ration the rumen makes little methane, but the grain carries its own footprint from the field. Beside it, four cards give drinking water, the feed's water footprint, the land to grow it and the area to spread the manure.

6.95 kg CO₂e per head per day

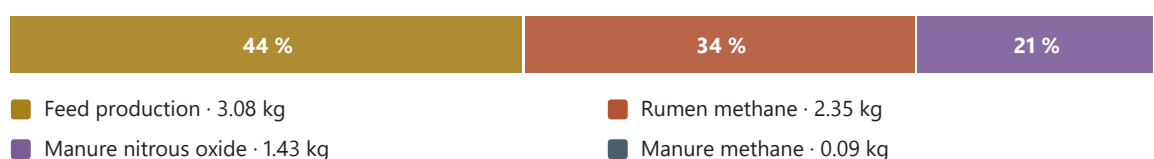


FIGURE 10.2 Where the gases come from. On a high-grain ration, the field that grew the feed weighs more than the rumen. Data from the case.

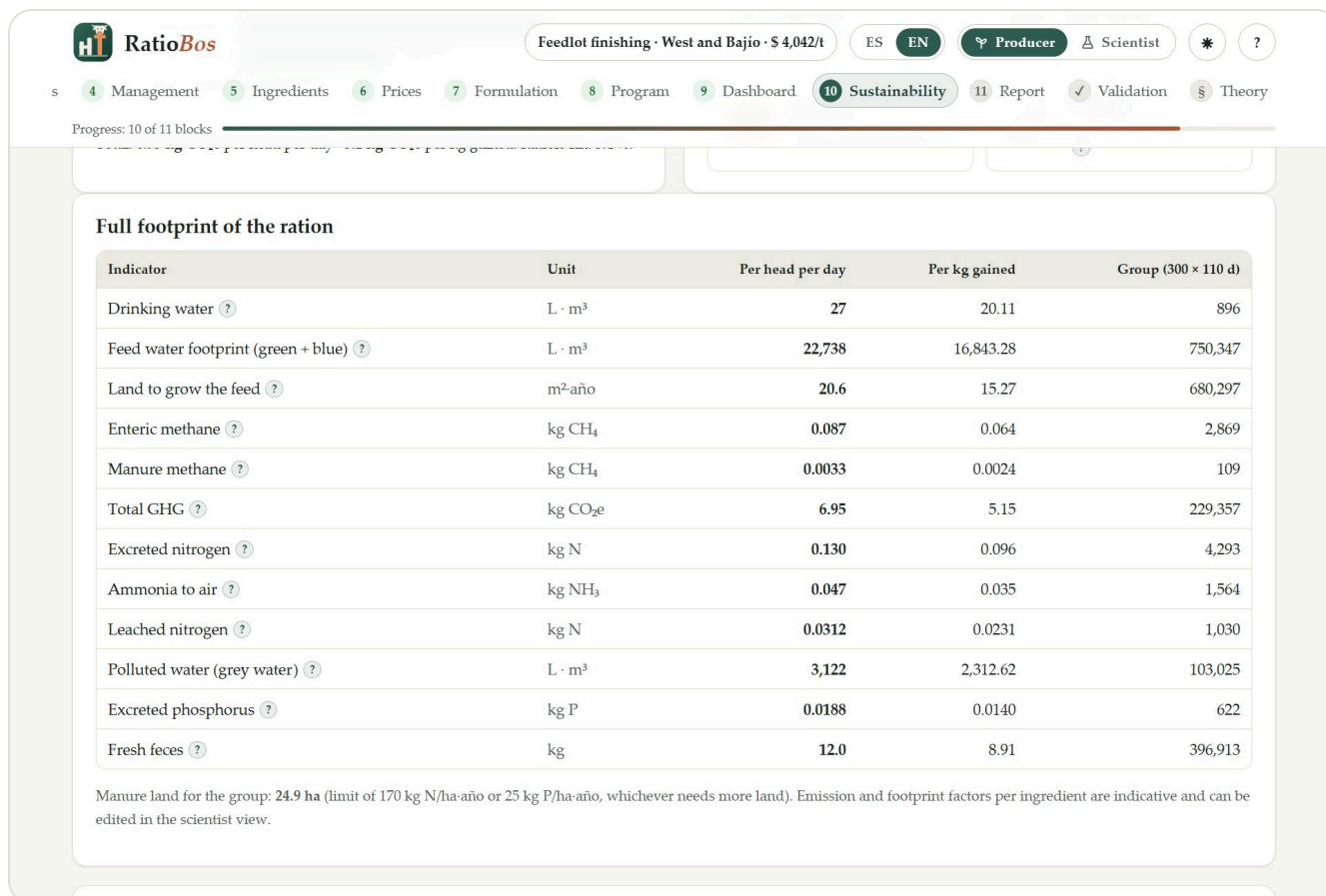


FIGURE 10.3 Full footprint per head per day, per kg gained and for the whole lot.



IN THE PEN

Our 300 bulls, over 110 days, produce about **397 tonnes of fresh faeces** with 4.3 t of nitrogen and 0.6 t of phosphorus. To spread it without polluting takes **24.9 hectares** of cropland (limit of 170 kg N or 25 kg P per hectare a year). If you do not have them, compost the manure or sell it to someone who does.

10.3 Medicines by category

From what you entered in Block 4, the app works out how much of each product each animal and the whole lot receive, and classifies it: ionophores (no importance in human medicine), hormonal implants, antiparasitics, vaccines and, if used, antibiotics by WHO category and β -agonists.

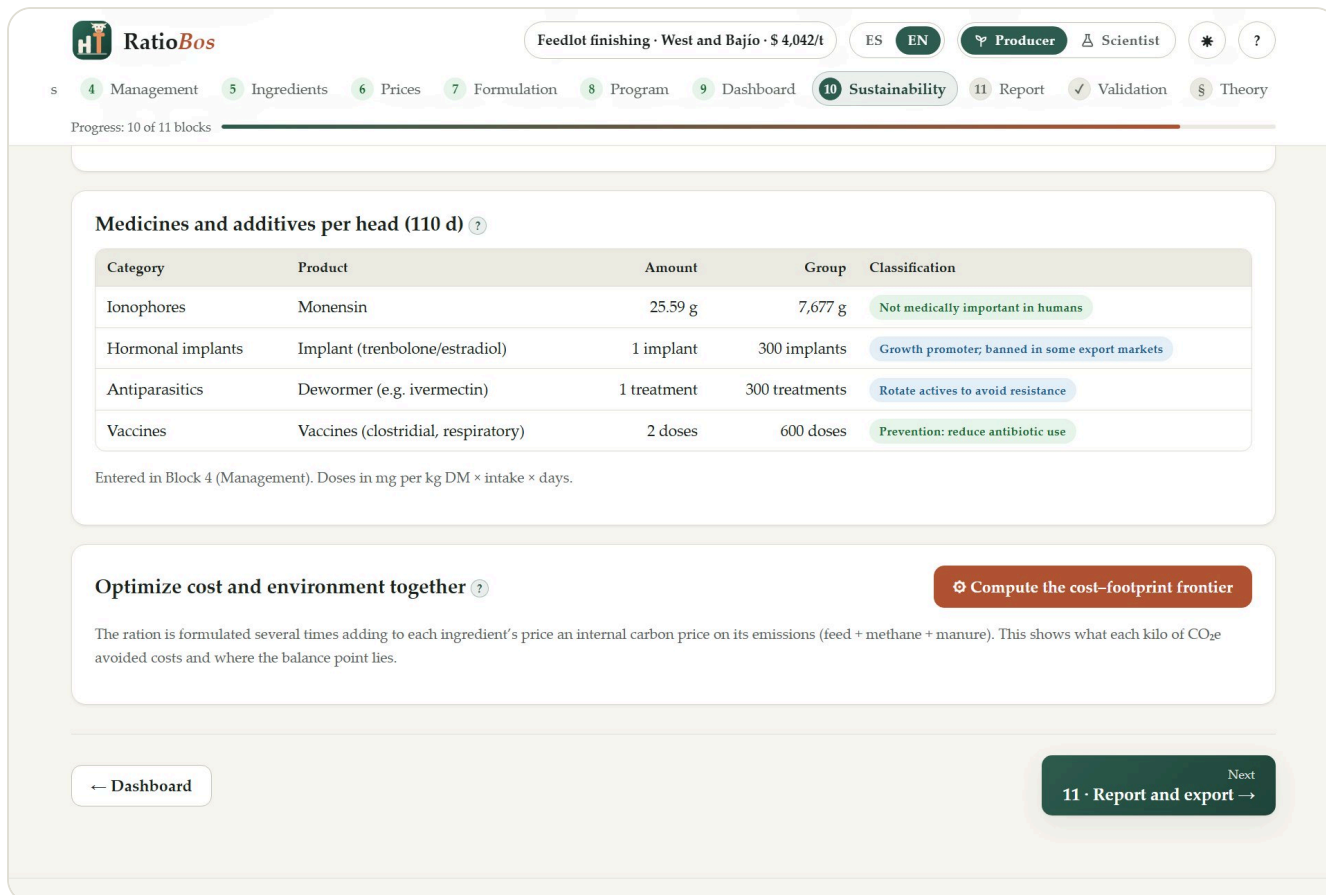


FIGURE 10.4 Medicines and additives per animal, 110 days.

10.4 Optimising cost and environment together

The [Compute the cost-footprint frontier](#) button formulates several times, adding to each ingredient's price an **internal carbon price** for its emissions, from \$0 to \$20,000 per tonne of CO₂e. The result is a curve of the best possible rations; no other is both cheaper and cleaner. The **balance** point is the one that removes the most footprint per extra peso.

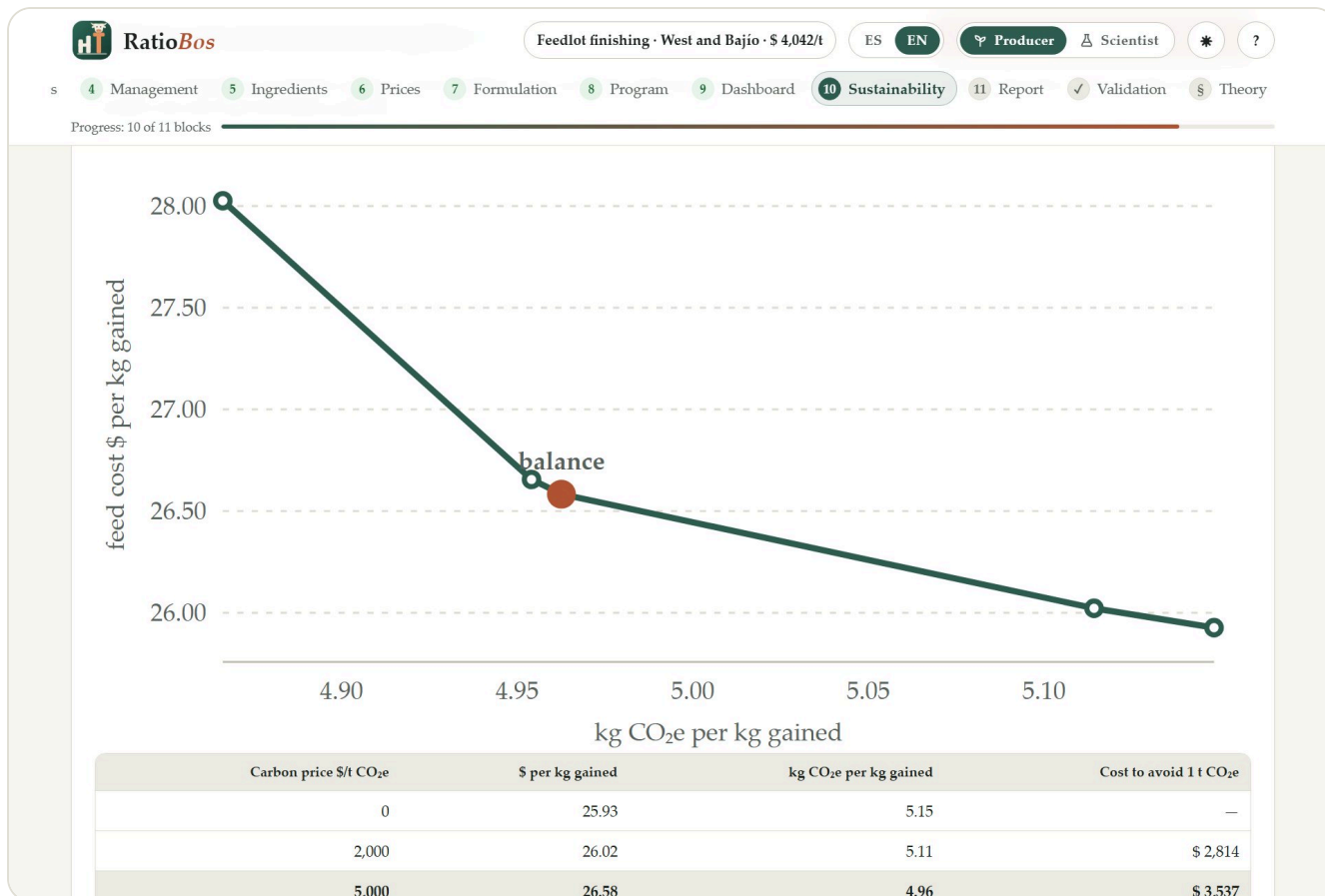


FIGURE 10.5 Cost-footprint frontier.

1

Use the balance ration sends it to Block 7 as your ration.

LEAST COST

\$ 25.93 · 5.15 kg



BALANCE · \$5,000/T CO₂E

\$ 26.58 · 4.96 kg



IN THE PEN

With an internal price of \$5,000 per tonne, the app swaps part of the sorghum for molasses and DDGS: the footprint falls 3.6 % (from 5.15 to 4.96 kg CO₂e per kilo gained) and each kilo gained costs 2.5 % more (\$26.58). Each tonne of CO₂e avoided costs about \$3,500, in the range of the social cost of carbon. It is a business decision: worth it if your buyer pays a premium for lower-footprint beef.

11

Report and export



A report for your advisor, tables for Excel and a file with your whole project.

11.1 The report

A one-sentence summary and the formula on a dry-matter basis, as fed and in kilos per tonne of mix, with the pen, ranch and date. [Print / PDF](#) prints it or saves it as a PDF to send by WhatsApp or e-mail.

Feedlot finishing · West and Bajío · \$ 4,042/t
ES EN
🔍 Producer
🔍 Scientist
⚙️
?

ts
4 Management
5 Ingredients
6 Prices
7 Formulation
8 Program
9 Dashboard
10 Sustainability
11 Report
✓ Validation
💰 Theory

Progress: 11 of 11 blocks

Feedlot · pen 4

Example ranch, Jalisco · 10/6/2026 · RatioBos 1.0.0

Ration for feedlot finishing · zebu × european cross · bull weighing 380 kg. It costs \$ 4,042 per tonne as mixed; each animal eats 8.7 kg per day at \$ 35.00 daily. Expected gain is 1.35 kg per day, at \$ 25.93 of feed per kilo gained. RatioBos computed it to be the cheapest possible with your ingredients and prices. Review it with your veterinarian or nutritionist before feeding and change rations gradually.

Ingredient	% MS	% AF	kg AF per t of mix
Dry-rolled sorghum	83.76	83.35	833.5
Corn stover	11.20	11.80	118.0
Distillers grains with solubles (DDGS)	2.80	2.82	28.2
Calcium carbonate	0.92	0.83	8.3
Urea (46 % N)	0.81	0.73	7.3
Trace-mineral and vitamin premix	0.30	0.28	2.8
Salt (NaCl)	0.20	0.18	1.8

Export

[Print / PDF](#)
[↓ Formula \(CSV\)](#)
[↓ Nutrients \(CSV\)](#)

Project

FIGURE 11.1 Ration report.

11.2 Export and save

BUTTON	WHAT IT DOES
 Formula (CSV)	The formula with prices and cost share, for Excel.
 Nutrients (CSV)	Each nutrient's supply against its minimum and maximum.
 Save project (.json)	Everything: animal, region, ingredients, prices, result and program. Open it on another computer with  Open project .
Reset everything	Back to factory values. Save your project first.



TIP · ONE FILE PER PEN AND DATE

Name the project with the pen and date (for example, "Pen 4 · 2026-10-06"). When the feeding period ends you can compare what the app predicted with what your animals weighed: the best way to fine-tune your ranch's data.

From paper to feed bunk

A

The best ration is worthless if it is mixed badly or changed overnight. Five minutes of checking prevent a wrong load and an acidosis outbreak.

A.1 Before loading the wagon

- ☐ The **prices** in Block 6 are this week's, delivered to the ranch.
- ☐ The silage **dry matter** is up to date: it changes with every silo and with rain. A microwave or oven test is enough.
- ☐ The **weight** and **target** in Block 3 are those of the lot that will eat this ration.
- ☐ Every ingredient in the formula is **in stock**, in the amount needed.
- ☐ The **wagon load** in Block 4 matches your mixer, and the scale is calibrated.
- ☐ Urea, salt, minerals and premix are **pre-blended** with a little ground grain.
- ☐ Your **veterinarian** or nutritionist has reviewed the report. RatioBos is a support tool, not a substitute.

A.2 Change rations gradually

The rumen needs time to shift its microbes from forage to grain. That is why the program starts with a receiving ration of 30–45 % forage. Between one ration and the next, switch over **3 to 5 days**, blending both or raising grain step by step, and never raise grain by more than 10 points at once.

! CAUTION · SIGNS OF ACIDOSIS

Loose, bubbly faeces, animals going off feed, lameness a few days later and intake that swings up and down. If you see them, go back to the previous ration and call your veterinarian.

A.3 Read the bunk every morning

Actual intake rules. Before the first feeding, look at what is left and adjust the day's amount:



DECISION RULE · BUNK READING

Licked clean

Raise the day's amount 5 %.

Crumbs, a thin layer

Right. No change.

5–10 % left

Cut 5 %. Check for heat or a sick animal.

Over 10 % left

Cut 10 % and check the mix, water and pen health.



TIP · CLOSE THE LOOP

Once you know your lot's actual intake over a week or two, enter it in Block 3 (**Actual DM intake**). The app will use it instead of its prediction, and the expected gain will match your pen.

B

Troubleshooting

The most common questions, with cause and fix.

I changed a price and the result disappeared.

“No solution” appears.

The expected gain is higher than my target.

The ration leaves out my main ingredient.

The wagon kilos are not what I expected.

I opened the app on another computer and my data is missing.

The THI says “emergency” but my cattle are fine.

Can I formulate for cows on rangeland?



Glossary

The technical words in this guide, in plain language.

Rumen acidosis <i>Acidosis ruminal</i>	Too much acid in the rumen from much grain and little fibre; cuts intake and causes lameness.
Dry basis (DM) <i>Base seca (MS)</i>	Feed without its water. Ingredients are compared and the ration computed this way.
Benefit : cost <i>Beneficio : costo</i>	Money in over money out. Above 1 means profit.
Biotype <i>Biotipo</i>	Group of breeds that behave alike: British, Continental, zebu, cross, Criollo, dairy.
Feed conversion <i>Conversión alimenticia</i>	Kilos of feed (DM) per kilo of weight gained. Lower is better.
CO₂ equivalent (CO₂e) <i>CO₂ equivalente (CO₂e)</i>	Gases added by their climate effect: 1 kg methane = 27 kg CO ₂ ; 1 kg nitrous oxide = 273.
Feed efficiency <i>Eficiencia alimenticia</i>	Kilos gained per kilo of feed. Higher is better.
Net energy for gain (NEg) <i>Energía neta de ganancia (ENg)</i>	The feed energy that becomes kilos, in Mcal per kg DM.
Net energy for maintenance (NEm) <i>Energía neta de mantenimiento (ENm)</i>	The energy the animal spends just staying alive, without gaining weight.
Physically effective NDF (peNDF) <i>FDN físicamente efectiva (FDNfe)</i>	Long fibre that makes cattle chew cud and produce saliva, the rumen's natural buffer.
ADG <i>GDP</i>	Average daily gain, in kg per day.

Ionophore <i>Ionóforo</i>	Additive (monensin, lasalocid) that improves energy use and reduces acidosis.
THI <i>ITH</i>	Temperature-humidity index: the heat the animal feels.
Least cost <i>Mínimo costo</i>	The cheapest ration among all those that meet the requirements.
Entry price <i>Precio de entrada</i>	Price at which an ingredient left out would start to pay off.
Break-even price <i>Precio de equilibrio</i>	Sale price per kilo live weight at which the lot neither gains nor loses.
Premix <i>Premezcla</i>	A pre-blend of the small ingredients with a little ground grain, so they spread evenly.
Metabolizable protein (MP) <i>Proteína metabolizable (PM)</i>	The protein that actually reaches the gut: rumen microbial protein plus undegraded protein.
Sustainability <i>Sostenibilidad</i>	As FAO defines it: producing today without compromising the capacity to produce tomorrow, environmentally, economically and socially.
As fed (AF) <i>Tal como se ofrece (TCO)</i>	Feed with its water, as the wagon scale weighs it.
Urea <i>Urea</i>	Cheap non-protein nitrogen that rumen microbes turn into protein. At most $\approx 1\%$ of DM.

RatioBos on one page

hectortecumshe-ai.github.io/RatioBos

Scan to open the app



1

Animal

Animal type, biotype (cross, zebu...) and sex.

3

Requirements

Lot weight and daily gain target.

5

Ingredients

Tick what you have. Fix the silage DM.

7

Formulation

⚙ Formulate. Check the traffic light and print the wagon sheet.

9

Dashboard

ADG, efficiency, conversion, B : C and the optimal gain.

11

Report

PDF for your advisor and Save project (.json).

2

Region

Tap your state and season. Check THI, shade and floor.

4

Management

Implant, ionophore, head count, wagon load, buy and sell.

6

Prices

This week's delivered price. Check the maximums.

8

Program

⚙ Formulate the program: days on feed and profit per head.

10

Sustainability

Ration footprint and cost-footprint frontier.

IF "NO SOLUTION"

- Use the maximum gain it proposes.
- Recompute the Block 3 table.
- Tick the missing ingredient.

BEFORE MIXING

- Prices and silage DM up to date.
- Urea and minerals pre-blended.
- Ration change over 3–5 days.

RatioBos · Producer Guide, 1st edition, October 2026. Covers RatioBos 1.0.0. Every screenshot comes from the real app with the case data in section 0.5; prices are illustrative.

Decision-support tool: results should be reviewed by a qualified animal nutritionist. © 2026 Héctor Tecumshé Mojica-Zárate. The software is distributed under the MIT License.